



Ensemble 6000 Series LDS6525 Low Density Server Device Guide

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Preface

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WARNING: EXPORT CONTROLLED. These commodities, technology, or software are controlled for export from the United States in accordance with the Export Administration Act of 1979 as amended (Title 50 U.S.C.; App. 2401, et seq.), through the Export Administration Regulations (EAR). Diversion contrary to U.S. law is prohibited.

About This Document

This device guide provides information on the Ensemble® 6000 Series OpenVPX™ Intel® LDS6525 Low Density Server Module (hereafter referred to as the LDS6525). This information includes, but is not limited to:

- Module descriptive overview
- Component descriptions
- Module topology
- Electrical, mechanical, and environmental specifications
- Signal and pin assignments
- Module management/configuration
- Installation procedures
- FRU and sensor data

Audience

Table 1 below lists the possible audience members for this document.

Table 1: Audience Members

Title	Description
System Managers	Establish and maintain development systems for application developers. Tasks include installation, configuration, testing, and system startup.
System Designers	Produce high-level designs for embedded hardware subsystems and software applications that are part of a use-domain system.
Application Designers	Produce high-level designs for multicomputer software applications.
Application Developers	Develop software programs for multicomputer applications. Tasks include program design, coding, debugging, and optimization.
System Integrators	Assemble and verify Mercury and non-Mercury multicomputer system hardware or software.

This guide assumes familiarity with the following technologies or concepts:

- Embedded computing
- VITA 46 and VITA 65/OpenVPX™ standards
- RapidIO® switch fabrics
- Ethernet communications
- FPGA IP development
- Linux® or VxWorks® operating systems
- C/C++ programming and debugging techniques
- Cross development tools, and the Eclipse™ development environment

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Related Documents

This publication might refer to other Mercury publications or to third-party documentation. For quick reference, Mercury suggests that you obtain copies of these documents.

Mercury Documents

Mercury documents are provided with purchased products. Table 2 below lists the Mercury documentation that is related to this product.

Table 2: Related Mercury Documentation

Title	Location
<i>MultiCore Plus® DIAGSDK Diagnostics Guide</i>	http://www.mrcy.com/
<i>Ensemble 6000 Series LDS6525 Low Density Configuration Guide</i>	http://www.mrcy.com/

During software installation, Mercury software documentation is installed in the /mercury/doc folder.

Third-Party Documents

Table 3 below lists the third-party documentation that is relevant to this product.

Table 3: Relevant Third-Party Documentation

Document	Description
MIL-STD-810	Department of Defense Test Method Standard for Environmental Engineering Considerations and Laboratory Tests http://www.dtc.army.mil/publications/milstd.aspx Establishes uniform methods for environmental test for determining the resistance of aerospace and ground equipment to the deleterious effects of natural and induced environments peculiar to military operations.
MIL-HDBK-217F	Military Handbook: Reliability Prediction of Electronic Equipment Available from several sources through online search. Describes consistent and uniform methods for estimating the inherent reliability of military equipment and systems.
MIL-HDBK-454A	Military Handbook: Safety Design Criteria - Personnel Hazards

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Document	Description
	Available from several sources through online search. Guideline 1, Paragraph 4.9 - Hazardous and restricted materials.
MIL-HDBK-5400	<i>Military Handbook: Electronic Equipment, Airborne, General Guidelines for</i> Available from several sources through online search. Contains general guidelines for electronic equipment for operation in piloted aircraft and helicopters, missiles, boosters and allied vehicles.
Telcordia SR-332-1	<i>Reliability Prediction Procedure for Electronic Equipment</i> http://telecom-info.telcordia.com Describes recommended methods for predicting device and unit hardware reliability.
ANSI/VITA 47-2005 (R2007)	<i>Environmental, Design and Construction, Safety, and Quality for Plug-In Units Standard (Vibration Class 3)</i> http://www.vita.com/specifications.html Defines environmental, design and construction, safety, and quality requirements for commercial-off-the-shelf (COTS) plug-in units (cards, modules, etc) intended for mobile applications.
VITA 46.0	VPX http://www.vita.com/specifications.html Defines an open standard for supporting high speed, switched interconnect protocols on 3U and 6U IEEE1101 form factors with a new connector scheme.
VITA 46.2	<i>VPX PCI Bus Signal Mapping</i> http://www.vita.com/specifications.html Defines mapping of PCI bus signals to a VITA 46 connector
VITA 48.0	<i>REDI (Ruggedized Enhanced Design Implementation)</i> http://www.vita.com/specifications.html Defines a general mechanical design implementation for circuit card assemblies that enhance thermal performance, structural integrity, and 2-level maintenance compatibility.
VITA 48.1	<i>Mechanical Specifications for Microcomputers Using REDI Air Cooling Applied to VITA 46</i> http://www.vita.com/specifications.html Defines the critical dimensions to ensure mechanical interchangeability of air cooled VITA 46 plug-in modules.
VITA 65.0	OpenVPX http://www.vita.com/openvpx/

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Document	Description
	Describes the technical implementation details for 3U and 6U VPX payload and switch modules, backplane topologies, and chassis products, and provides guidance on how to build interoperable computing and communication platforms.
Intelligent Platform Management Interface Specification v1.5	
	http://www.intel.com/design/servers/ipmi/ The base specifications for the Intelligent Platform Management Interface (IPMI) architecture; defines standardized, abstracted interfaces to the platform management subsystem. Includes the definition of interfaces for extending platform management between boards within the main chassis, and between multiple chassis.

Programming Examples

Any programming examples in this document might include long command lines. Regardless of its appearance in this document, each command must be on a single line. Individual commands appear separated by a line break.

Software Policy

The Mercury Software Release Policy is available online at <http://www.mrcy.com/>.

Conventions

This publication might use the following typographic, syntactic, and command line conventions.

Table 4: Typographic Conventions

Convention	Description
/s/node/setup.exe	Function, file, and path names
argument_value	User-specified value
MC_ENV	Environment variables
Ctrl+Z, Open File dialog box	Keystrokes and user interface elements
Start > Settings	User interface menu command selection

Table 5: Syntactic Conventions

Convention	Description
{ }	Indicates a required element. In this group, select one element only
[]	Indicates an optional element
{choice choice}	Indicates that you must choose one item
[choice choice]	Indicates that you can choose either item
... [, ...] [; ...]	Indicates that the preceding syntax element can be repeated any number of times, with a separator between elements
0x200	Represents a hexadecimal value
0b1001	Represents a binary value
Ellipsis (...)	Indicates omitted text in a list or example

Table 6: Command Line Conventions

Convention	Description
/s/node/setup.exe	Function, file, and path names
#	Linux® root prompt
%	Linux user prompt
->	VxWorks® operating system prompt
C:\>	Microsoft® command prompt
C:\> remcall -x wr_32	Interactive input on a command line

Chapter 1

LDS6525 Overview

LDS6525 Introduction

The LDS6525 is a single-processor module in a 6U, 1" pitch, VITA 48.0 and OpenVPX VITA 65 form factor with one PMC/XMC site, and one XMC only site. The LDS6525 includes a single Compute Reuse Block (CRB); the CRB includes an Intel quad core Haswell mobile processor, 8 GBytes DDR3 memory and a Lynx Point mobile platform controller hub (PCH). For a full summary of module features, see [LDS6525 Features](#) on page 3.

The LDS6525 is available in the following versions:

Variant	Mercury Part Number	Description
Air-cooled	910-56112-01	L0 air cooled variant.
Air-cooled	910-56112-09	L0 air-cooled P-FPGA-enabled variant.

[Figure 1-1](#) on the next page shows the air-cooled LDS6525

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Figure 1-1: The Air-Cooled LDS6525.

 CAUTION	The LDS6525 is not hot-swappable. You must remove all power before removing the module from, or inserting the module into a chassis.
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LDS6525 Features

Table 1-1 below summarizes the LDS6525 features.

Table 1-1: LDS6525 Feature Summary

Feature	Description
Processor	■ Intel Core i7 processor ■ Two memory controllers (64 bits of data) ■ 8 GB DDR3 SDRAM (4 GB per controller) ■ 8 GB NAND Flash via SATA interface ■ Communication with the PCH through: ■ Direct Media Interface (DMI) ■ Intel Flexible Display Interconnect (IFDI) ■ Platform Environmental Control Interface (PECI) ■ Realtime clock
Ethernet interfaces	■ x4 PCIe® Gen 2 between the PCH and the 82580 Quad Ethernet Controller ■ Dual backplane 1000BASE - BX interface ■ Dual backplane 10/100/1000BASE-T through magnetics
USB Interfaces	■ Three USB 2.0 interfaces to backplane ■ One USB 2.0 interface to front panel (air-cooled only) ■ Three USB 3.0 ports to the backplane
PCI Express® Interfaces	■ One x16 PCIe Gen 3 interface to processor ■ One x8 PCIe interface per XMC site ■ Two x8 PCIe interfaces to FB-FPGA ■ Two x8 PCIe interfaces to the P2 backplane connector ■ One x8 PCIe interface to the P5 backplane connector ■ One x4 PCIe interface to Tsi384 PCI-X bridge to the PMC site ■ One x1 PCIe interface to each XMC site for FPGA configuration

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Feature	Description
Platform Controller Hub (PCH)	Mobile Intel 7 Series Chipset ■ Lynx Point Mobile QM87 platform controller hub ■ PCIe interface to the Ethernet controller
UART Interfaces	■ Four UART connections to the backplane, accessible by RTM or backplane interface module ■ Two ports support RS-232/RS-422/RS-485, selected through a soft jumper setting ■ Two ports support RS-232 only ■ Two RS-232 connections to a front panel micro DB-9 connector (air-cooled only)
Fabric Bridge	■ Two PCIe x8 Gen3 interfaces ■ Four x4 data plane ports ■ Ethernet support: ■ 10GE support ■ 40G-KR (x4 10.3125 Gbps 64b/66b) support ■ Four PCIe ports of 4 lanes (x4) support ■ Four SRIO ports of 4 lanes (x4) support ■ Four front panel status LEDs ■ 35mm x 35mm FCBGA package ■ NOR Flash support
FPGAs	■ An M-FPGA that provides the management engine for the module ■ A P-FPGA that supports interfaces to PCH, M-FPGA and TPM; sensor monitoring; and Non-volatile memory gate-keeping logic ■ An FB-FPGA that supports bridging functionality between the PCIe switch and the backplane data plane signals
Trusted Platform Module	■ Intel® AT97SC3204 security module that conforms to Trusted Computing Group (TCG) v1.2 specifications
PMC/XMC Sites	■ One PMC/XMC site and one XMC only site with: ■ I²C interface to the management processor (M-FPGA) ■ One x8 PCIe port routed to the PCIe switch ■ One x1 PCIe port for FPGA configuration

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Feature	Description
System Management	■ On-board IPMI controller ■ Voltage/temperature monitor ■ FRU and on-board EEPROM interfaces
Power Subsystem	■ Power managed by M-FPGA ■ ECB protection on 12V power input ■ Current limiting on 3.3V Aux power input
Test and Debug	■ P-FPGA, M-FPGA UART interface to front panel UART connector ■ JTAG boundary scan interface
Open VPX Backplane	■ Data Plane: four x4 lanes ■ Four Ethernet 40G-KR ports (x4 10.3125Gbps 64b/66b) possible ■ Four SRIO ports of 4 lanes possible ■ Four PCIe ports of 4 lanes possible ■ Control Plane: two 1000BASE-BX ports ■ Management Plane: redundant I²C from P0 backplane P0 to M-FPGA ■ Expansion Plane: two x8 PCIe interfaces to the P2 connector

LDS6525 Topology

Figure 1-2 below shows a block diagram of the LDS6525 topology.

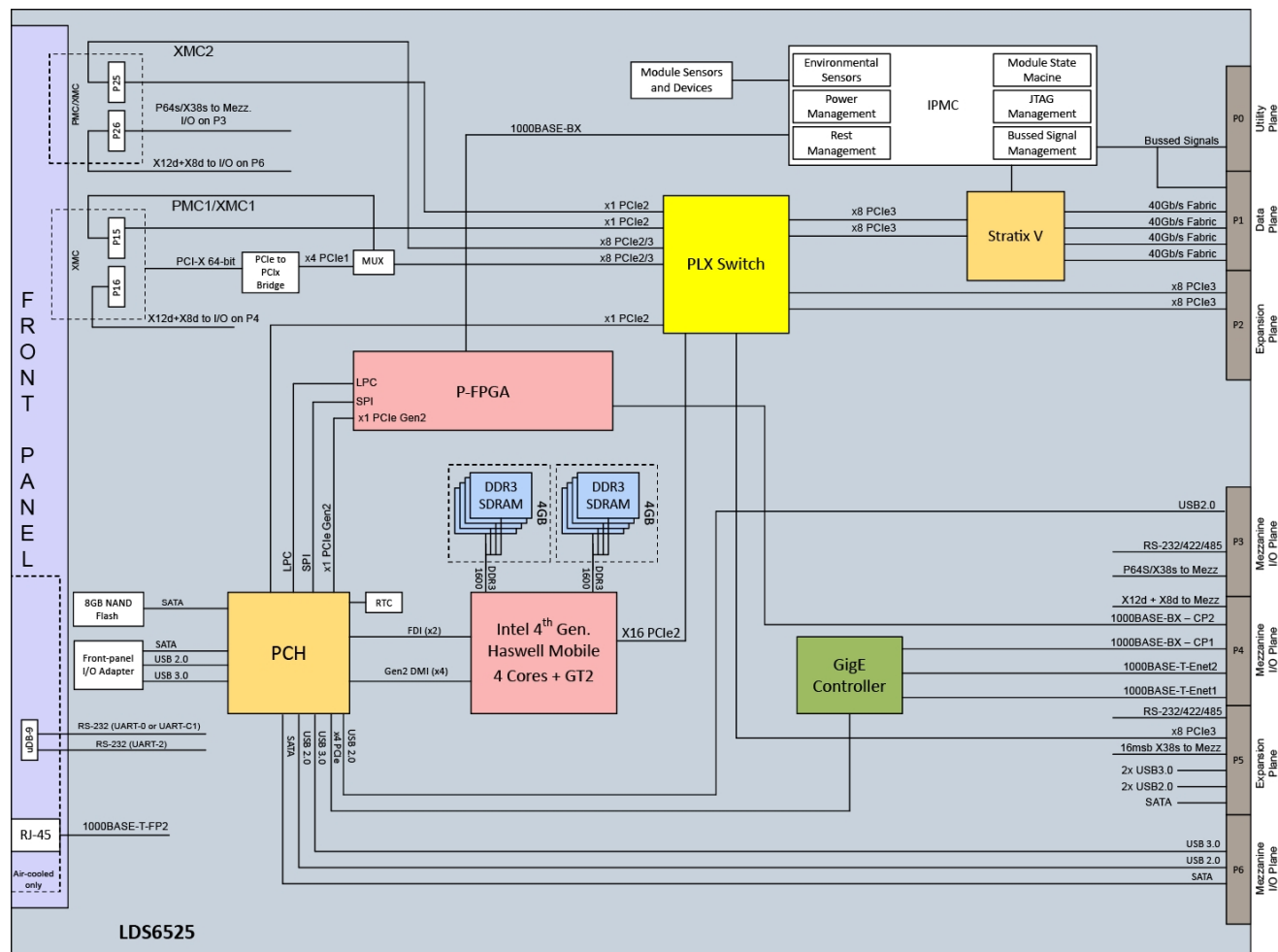


Figure 1-2: The LDS6525 Module Topology

Compatibility

The LDS6525 is compatible with the Mercury and third-party products described herein.

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Chassis Compatibility

As defined by the VITA 48.1 specification, the air-cooled LDS6525 is a 1-inch-pitch Type II plug-in unit (does not meet Level 2 maintenance requirements) that is intended for use in a VITA 48.1-compliant air-cooled chassis, such as the Ensemble 6000 chassis.

Compliance

The LDS6525 adheres to the following standards and restrictions. For quick reference, Mercury recommends that you obtain copies of these specifications.

Hazardous or Restricted Materials Compliance

The components in the air-cooled LDS6525 comply with RoHS 5/6, which restricts the use of hazardous materials in the manufacture of various types of electronic and electrical equipment.

The LDS6525 module is designed to not contain hazardous or restricted materials as defined by MIL-HDBK-454A, Guideline 1, Paragraph 4.9.

The LDS6525 might contain one or more of the banned RoHS substances.

Reference Standards

The LDS6525 complies with the following standards:

- ANSI/VITA 42.0 *Standard for VITA 42.0 XMC*
- ANSI/VITA 42.3 XMC PCI Express Protocol Layer Standard
- ANSI/VITA 46.0-2007 *VPX Base Specification*
- ANSI/VITA 46.3 *Serial RapidIO Signal Mapping for VITA 46 (VPX)*
- ANSI/VITA 46.4 *PCI Express on VPX Fabric Connector*
- ANSI/VITA 46.7 *Ethernet on VPX Fabric Connector*
- ANSI/VITA 46.9 *PMC/XMC/Ethernet Signal Mapping to 3U/6U on VPX User IO Draft Standard*
- ANSI/VITA 46.10 *Rear Transition Module for VPX*
- ANSI/VITA 46.11 *System Management on VPX*
- ANSI/VITA 48.0 *REDI (Ruggedized Enhanced Design Implementation)*
- ANSI/VITA 48.1 *Mechanical Specifications for Microcomputers Using REDI Air Cooling Applied to VITA 46*
- VITA 48.1 *Type 2, 1" Pitch*

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- ANSI/VITA 61.0 *XMC 2.0 Enhanced for Ruggedized and High-Speed*
- ANSI/VITA 65.0 *OpenVPX Base Specification*, slot profiles SLT6-PAY-4F1Q2U2T-10.2.1, SLT6-PAY-4F1Q2U2T-10.2.6
- IPMI software standard: *Intelligent Platform Management Interface Specification v1.5*

Chapter 2

LDS6525 Components

Front Panel Components

The air-cooled LDS6525 front panel provides status indicators, connectors, and a reset switch. [Figure 2-1](#) below shows the front panel components.

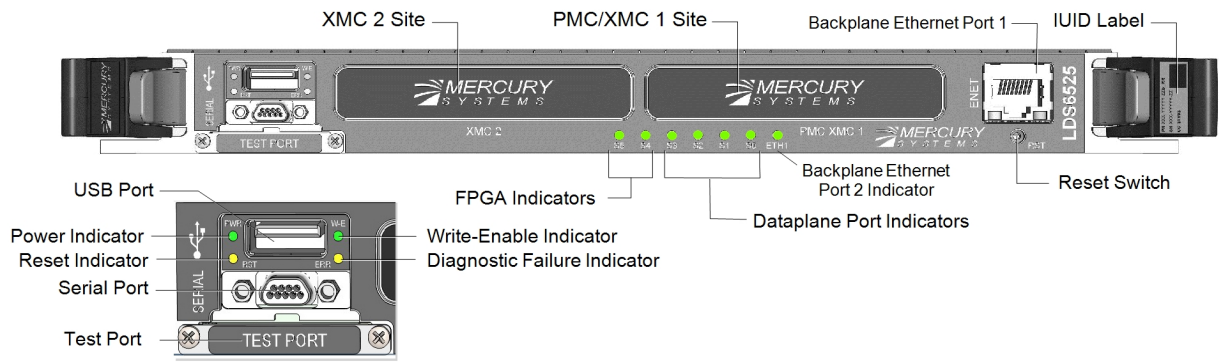


Figure 2-1: Air-cooled LDS6525 front panel components

Module Status Indicators

[Table 2-1](#) on the next page lists the LDS6525 front panel module status indicators. For the location of each indicator, see [Figure 2-1](#) above.

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Table 2-1: LDS6525 Front Panel Status Indicators

Label	Color	Function	Default State	Condition	Meaning	Contact Support If:
PWR	Green	Power good	On	On	Local module power planes are operating within specifications.	The PWR indicator does not turn on when you apply power.
				Flashing	Indicates a module power failure.	
				Off	Local module power supplies or backplane power rails are turned off.	
RST	Amber	Module reset	Off	Flashing	Module is in reset.	The module will not reset.
				Off	Module is not in reset.	
W-E	Green	Module write-enable	On	On	Software can modify the accessible non-volatile memory components on the module.	N/A
				Off	Software cannot modify the accessible non-volatile memory components on the module.	
ERR	Amber	Diagnostics failure	Off	On	A failure has been detected on the module, or the module is powered-on or in reset and on-board diagnostics are evaluating the module operational status.	The ERR indicator turns on and stays on.
				Off	The module is operating normally; no failure has been detected.	
S0 through S3	Green	FB-FPGA status	User-definable. Controlled by FB-FPGA bitstream.			
S4 and S5	Green	P-FPGA status	User-definable. Controlled by P-FPGA bitstream.			
ETH1	Green	LLD11 Activity	On	On	The Activity LED for the second backplane Ethernet port.	N/A

Certain indicator combinations have specific meanings. [Table 2-2](#) on the facing page lists the meaning of these combinations.

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Table 2-2: Front Panel Status Indicator Combinations

Label	Color	Condition	Meaning
PWR	Green	Flashing	This combination indicates that the power manager has detected a power failure.
ERR	Amber	On	
RST	Amber	On	
PWR	Green	Flashing	This combination indicates that the power supplies did not ramp up within 500 milliseconds after being commanded to turn on.
ERR	Amber	On	
RST	Amber	Off	

Ethernet Status Indicators

Air-cooled modules have front panel Ethernet connectors with integrated status indicators. [Figure 2-2](#) below shows the location of the indicators on the Ethernet connector.

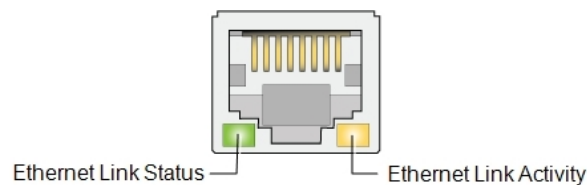


Figure 2-2: Ethernet indicator locations

[Table 2-3](#) below lists the functions of the Ethernet indicators.

Table 2-3: Ethernet Indicator Purpose

Purpose	Color	Description	Default State*	Action
Ethernet link activity	Amber	On: Data is being exchanged over this port Off: Data is not being exchanged over this port	Off	If there is no Ethernet activity, check your network connection.
Ethernet link status	Green	On: A 10BASE-T, 100BASE-T, or 1000BASE-T Ethernet connection is operational on this port. Off: The Ethernet port is not operational.	Off	If the indicator shows no change, check your network connection.

* Normal operation/power on.

Reset Switch

The LDS6525 has a front panel push-button reset switch that initiates a full reset of the module payload. To prevent accidental actuation, the press-and-release switch is recessed.

Front Panel Connectors

[Table 2-4](#) below lists the front panel connectors available on the air-cooled LDS6525.

Table 2-4: Air-Cooled LDS6525 Front Panel Connectors

Label	Function	Description
ENET	RJ-45 Ethernet port	10/100/1000BASE-T link to the Gigabit Ethernet switch
Serial	Micro DB-9 serial port	Provides two serial RS-232 links: ■ One connection to the PCH ■ One connection to the chassis manager
USB	USB 2.0 Port	Provides a USB interface to the PCH
Test Port	24-pin surface-mount header	Provides a customer-defined I/O interface to the P-FPGA

Notes

For the location of each connector, see [Figure 2-1](#) on page 9.

For information on connecting to the ports listed in [Table 2-4](#) above, see [Making Connections](#) on page 56.

For information on the pin assignments for these ports, see [LDS6525 Connections, Signals, and Pin Assignments](#) on page 29.

For information on board-level connectors, see [Board Level Components](#) on page 14.

Ethernet Ports

The air-cooled LDS6525 supports front panel and backplane-accessible Ethernet ports. The backplane ports include two 1000BASE-BX connections and two 10/100/100BASE-T interfaces. Port 0 of the 82580 connects, via a multiplexer, to the backplane Ethernet port ExtEnet2 or the Front Panel Ethernet Port 1. The multiplexer is controlled by the soft jumper, `FP_ETH_SELECTED`. For more information on this soft jumper, see For more information, see [Board Level Components](#) on page 14.

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There is one 10/100/1000BASE-T Ethernet connection dedicated to the backplane and two 1000BASE-BX connections to the backplane. [Table 2-5](#) below lists the Ethernet interfaces on the LDS6525 module.

Table 2-5: LDS6525 Ethernet Port Mapping

Intel 82580 Ethernet Controller Port	Type	Module Connectivity
Port 0	10/100/1000BASE-T	P4 Connector - ExtEnet2
Port 1	10/100/1000BASE-T	P4 Connector - ExtEnet1
Port 2	1000BASE-BX	P4 Connector - CP1
Port 3	1000BASE-BX	P4 Connector - CP2

Serial UART Ports

The air-cooled LDS6525 module provides six external UART ports.

- Two UART ports are provided by the front panel micro DB-9 connector
- Four UART ports are provided by the backplane

One of the front panel ports, Port 0, shares a UART connection with one of the backplane ports. This allows you to access the UART interface from either the front panel or the backplane.

For information on the pin assignments for this connector, see [Micro DB-9 Serial Connector Pin Assignments](#) on page 44.

USB Ports

The USB interface supports USB 2.0 and connects directly to port 5 of the PCH USB controller.

For information on the pin assignments for this connector, see [USB Connector Pin Assignments](#) on page 45

Test Port

The front panel test port is a 24-pin header, which provides a 16-bit single-ended I/O and allows you to access the P-FPGA. You can use the test port to debug your IP development, IP integration, and IP verification processes.

For information on the pin assignments for this connector, see [FPGA Test Port Connector Pin Assignments](#) on page 46.

Board Level Components

The board-level components on the LDS6525 include processors, FPGAs, jumpers, backplane connectors, sensors, and Non-volatile memory devices.

Figure 2-3 below shows the LDS6525 board-level component locations.

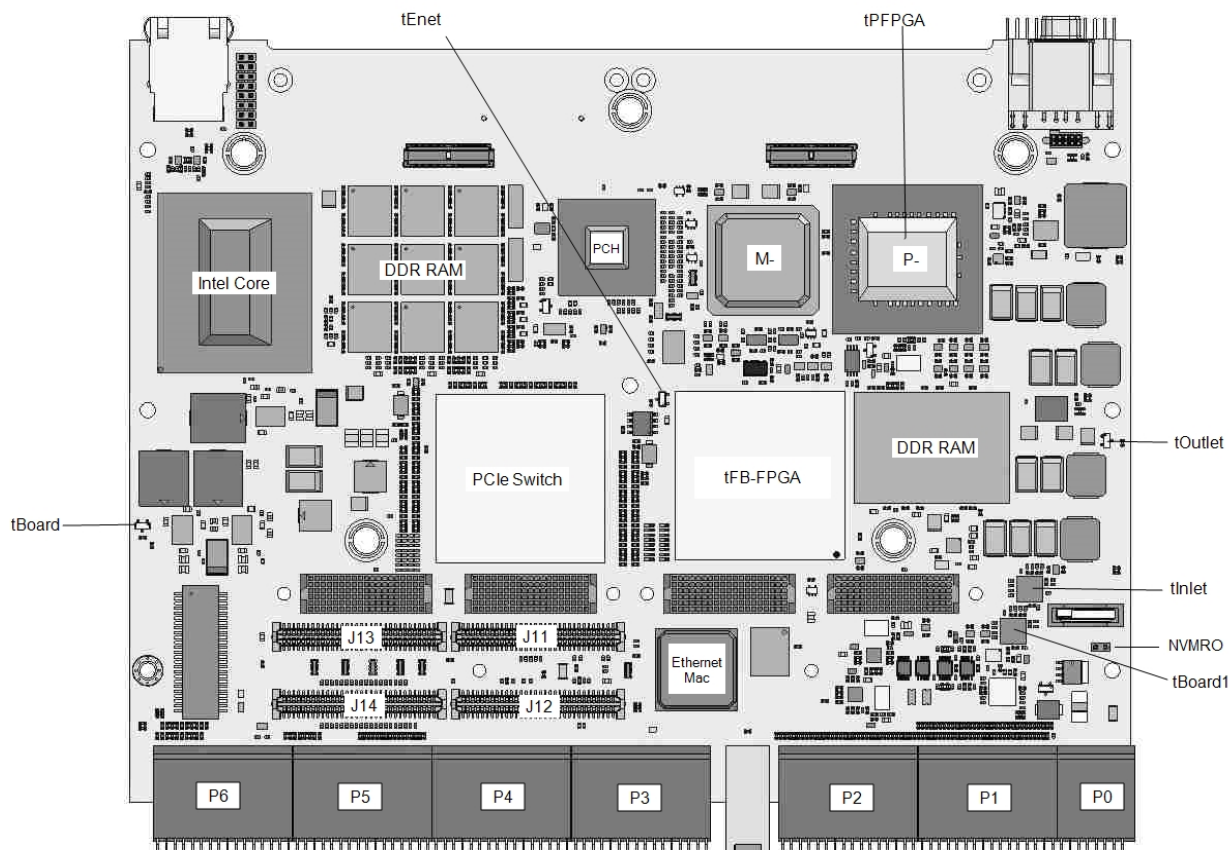


Figure 2-3: LDS6525 primary side board-level component locations.

Figure 2-4 on the facing page, provides a bottom view of the LDS6525 board.

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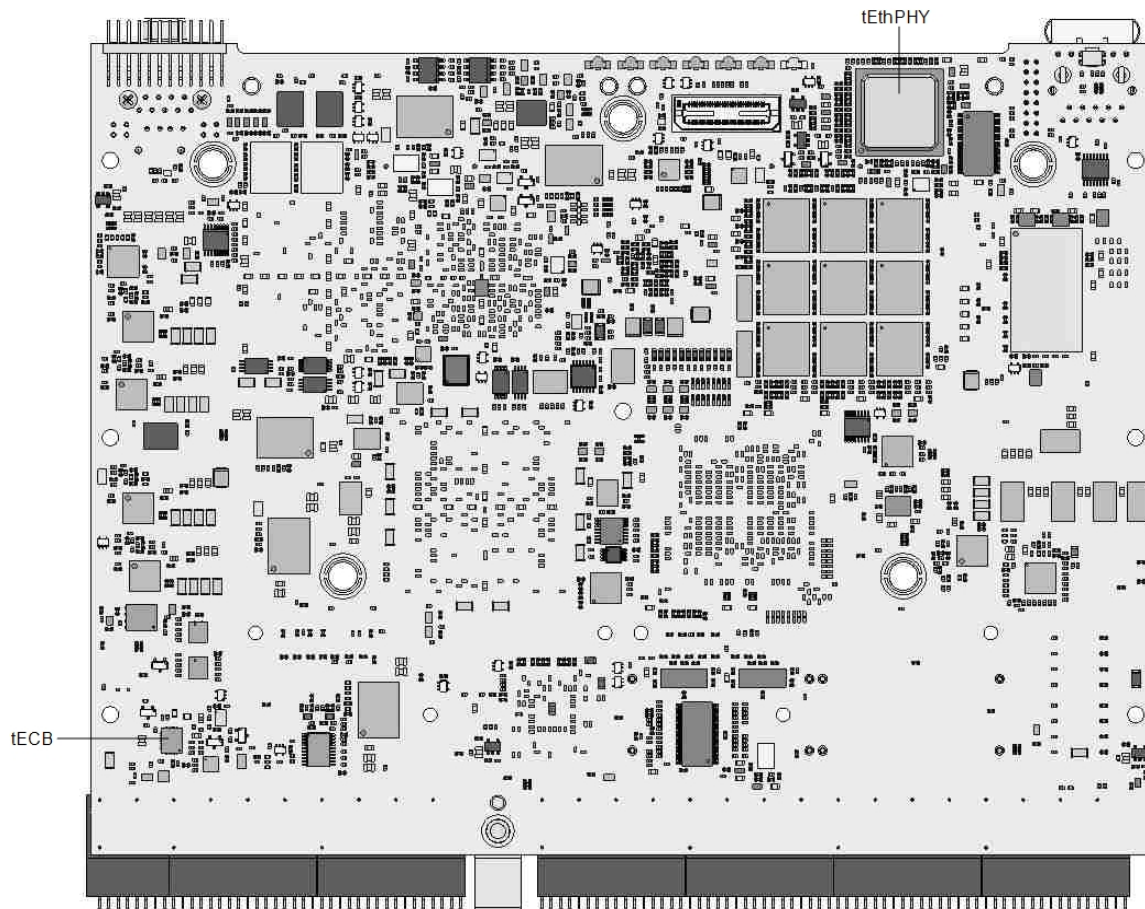


Figure 2-4: LDS6525 secondary side board-level component locations

Compute Processing Unit (CPU)

The LDS6525 CPU comprises an Intel Core i7 processor and a platform controller hub (PCH).

Intel Quad Core Mobile Processor

The Intel processor is a quad-core mobile-class device with two DDR3 SDRAM memory controllers. The processor communicates with the PCH via FDI, and DMI interfaces. The processor interfaces directly with the LDS6525 PCIe switches, and these switches use PCIe ports to interface with the XMC sites, the PMC site, the FB-FPGA, the P-FPGA, and the backplane expansion ports

Platform Controller Hub

The PCH is a Lynx Point Mobile Intel 7 Series Chipset Platform Controller Hub that supports two SPI flash devices and one NAND flash, accessible through a SATA interface.

The PCH supports USB and SATA interfaces at the front panel and the backplane. The PCH also interfaces with an Intel 82580 quad Ethernet controller through a x4 PCIe bus on ports 5 through 8

USB Support

The PCH supports eight USB 2.0 ports and six USB 3.0 ports. Six ports go to the backplane (three USB 2.0, three USB 3.0), and three to the front panel (air-cooled boards only) (two USB 2.0 and one USB 3.0). All of these ports are controlled using the Enhanced Host Controller Interfaces (EHCI) and the Extensible Host Controller Interface (xHCI).

SATA Support

The P6 backplane connector provides two SATA ports to the SATA controller on the PCH.

PCIe Support

The PCH PCIe interfaces to either the SATA, USB 3.0 or PCIe on ports 1 and 2. Ports 3 and 4 are unused, and ports 5-8 are used to interface with the 82580 Ethernet controller.

Ethernet Controller

The Ethernet controller is an Intel 82580 Quad Gigabit Ethernet mobile-class controller that supports four Ethernet ports through a PCIe connection to the PCH.

Fabric Bridge

The FB-FPGA provides a configurable serial I/O fabric bridge between the Haswell CRB and the backplane. By using an FPGA, rather than a dedicated component, the bridge functionality can be changed by loading a different FPGA personality image. In addition, the FPGA also provides a programmable block of logic that can be programmed by customers to meet their specific application needs. The device is packaged in a 35mm 1152 ball FCBGA. The estimated power for the FPGA is 16 Watts. The FB-FPGA has been provided with a maximum power budget of 15W for the core.

The FB-FPGA supports the following functions:

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- Fabric Bridge
- NOR Flash for configuration settings
- Processor GPIO
- SMBus interface
- InterFPGA interface

FPGA Devices

The LDS6525 has three FPGA devices, M-FPGA (Management FPGA), P-FPGA (Payload FPGA), and FB-FPGA (Fabric Bridge FPGA).

M-FPGA

The M-FPGA is a programmable device that provides the management engine for the module. It is accessible to the module's CPU, FB-FPGA and the P-FPGA. The M-FPGA supports the following functions:

- Power Management
- JTAG Multiplexer
- Module State Machine
- Reset Processing
- Non-volatile Memory Write-Protection
- Front Panel LED Control
- OpenVPX™ GDISCRETE Interface
- Backplane GPIO
- Control of OpenVPX™ Backplane Signals
- System Controller Functions
- LPC Bus Interface
- Intelligent Platform Management Controller (IPMC)
- Voltage, Current, and Temperature Sensor Interface
- IPMC Local module event logging; stored in NVM when module is not in the write protected state

Module State Machine

The M-FPGA2 contains the module state machine that is responsible for determining the operational state of the module, including:

- Power initialization
- Hardware load
- Reset
- Application ready (for operating)
- Failure conditions

To ensure that the module powers-up in a predictable manner, the module state machine interfaces with the power management logic, the IPMC, and the FPGA configuration logic, and recognizes the following states:

- Power off
- Power initialization
- Hardware load, reset, and fail

Reset Processing

The M-FPGA is responsible for driving all reset signals on the LDS6525 module. Reset outputs from the PCH are passed to the M-FPGA, which controls distribution to devices on the module.

Non-volatile Memory Write-Protection

The M-FPGA generates the write-protect signals for modifiable Non-volatile memory and non-modifiable Non-volatile memory. The module write-protection is governed by the write-protect jumper, NVMRO.

For more information on the modifiable and non-modifiable Non-volatile memory, see [Non-volatile Memory Devices](#) on page 23

Intelligent Platform Management Controller

The M-FPGA microcontroller sub-system (MSS) performs the function of an Intelligent Platform Management Controller (IPMC). The IPMC supports an Intelligent Platform Management Bus (IPMB) interface for communication to other system management modules in the chassis, a KCS interface to the CRB, and two UART interfaces:

- One UART interface is accessible from the front panel or the backplane, depending on a soft jumper setting.
- One UART interface is accessible by the CPU through the P-FPGA.

Sensors

The M-FPGA IPMC monitors the voltage, current, and temperature on the LDS6525 module. For more information, see [Temperature Monitoring](#) on page 63.

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P-FPGA

The P-FPGA is a programmable device that operates on the payload power rail and is accessible to the module's processor and management engine. The P-FPGA supports the following functions:

- LPC and PCIe interface with the PCH
- LPC arbitration logic
- GPIO interface with the M-FPGA and FB-FPGA
- FB-FPGA configuration control
- Processor and Backplane GPIO
- Interrupt Generation
- Processor Watchdog
- Global Clock Architecture
- Processor UART Interfaces (6 ports)
- I²C interface
- SATA NANDrive passthrough interface
- SRAM controller with interface SRAM and MRAM
- OpenVPX GDISCRETE interface
- Sensors
- Ethernet interface
- Non-volatile memory gatekeeping intercept logic for SPI FLASHs and EEPROM
- FPGA test port

FB-FPGA

The FB-FPGA provides bridging functionality between the PCIe switch and backplane data plane signals. The FB-FPGA supports the following functions:

- Fabric Bridge
- NOR Flash for configuration settings
- Processor GPIO
- SMBus interface
- InterFPGA interface

Trusted Platform Module

The Trusted Platform Module (TPM) on the LDS6525 module is a FIPS-1402-compliant random number generator that allows Trusted Computing Group (TCG)-compliant cryptography, including:

- RSA signature, verification, and key generation with an RSA modulo up to 2,176 bits
- Prime number generation up to 1088 bits for internal RSA key generation
- DSA signatures and verifications with parameters from 512 to 1,088 bits
- SHA-1 hash function

The TPM is connected to the PCH LPC bus through the P-FPGA. For details regarding how to use the TPM, see the Mercury technical note Using Trusted Boot on Mercury Intel Payload Modules located on the Mercury customer Web site at: <http://www.mrcy.com/>.

Ethernet Support

The Ethernet controller on the LDS6525 module is an Intel 82580 quad Gigabit Ethernet controller that supports four Ethernet ports (one x4 PCIe port running at 2.5Gbps) through a PCIe connection to the PCH.

The LDS6525 has front panel- and backplane-accessible Ethernet ports. The backplane ports include two 1000BASE-BX (SERDES) connections and one 10/100/100BASE-T (copper) interface.

Using the `FP_ETH_SELECTED` soft jumper, you can configure the 10/100/1000BASE-T Ethernet interface to connect to either the front panel port or the backplane. For more information, see *LDS6525 Soft Jumpers* on page 68.

Table 2-6 below shows the port mapping for the LDS6525 Ethernet interfaces.

Table 2-6: Ethernet Interface Port Mapping

Ethernet Controller Port	Type	Module Connectivity
Port 0	10/100/1000BASE-T	Front panel connector or P4 connector (ExtEnet2)
Port 1	10/100/1000BASE-T	P4 connector (ExtEnet1)
Port 2	1000BASE-BX	P4 connector (CP1)
Port 3 or P-FPGA	1000BASE-BX	P4 connector (CP2)

For information on the front panel RJ-45 connector and P4 backplane connector pin assignments, see *Connections, Signals, and Pin Assignments* on page 1.

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PCIe Switches

The PCIe switches on the LDS6525 module are 96-lane, 24-port PCI Express Gen 3 switches that provide connections to the Intel processor, the XMC sites, a PCI/PCI-X bridge, the backplane expansion plane, and the fabric bridge.

The switches provide:

- One x16 PCIe Gen 3 interface to the Intel processor
- One x8 PCIe interface per XMC site
- Two x8 PCIe interfaces to a FB-FPGA fabric bridge
- Two x8 PCIe interfaces to the P2 backplane connector
- One x4 PCIe interface to a PCI-X bridge to the PMC site
- One x8 PCIe interface to the P5 backplane connector

Module Sensors

The IPMC on the M-FPGA monitors the LDS6525 module voltages, current, and temperature. For more information, see [Temperature Monitoring](#) on page 63.

Jumpers

To change the behavior of some module functions, the LDS6525 provides mechanical (hard) and software (soft) jumpers.

For a complete list of the available soft jumpers and information on how to set them, see *LDS6525 Soft Jumpers* on page 68

About Mechanical Jumpers

The hard jumpers on the LDS6525 are provided to allow you to access certain features. Most hard jumpers are designed with 2-pin headers, so you can install or remove them with ordinary tools such as small pliers or tweezers.

Air-cooled module hard jumpers are accessible even when mezzanine cards are installed. [Figure 2-3](#) on page 14 shows hard jumper locations on the LDS6525.

The hard jumpers are logically configured as either “not jumpered” (no shunt installed) or “jumpered” (shunt installed). The default configuration for all hard jumpers is “not jumpered.”

[Figure 2-5](#) on the next page, below, shows the two-pin hard jumper settings.

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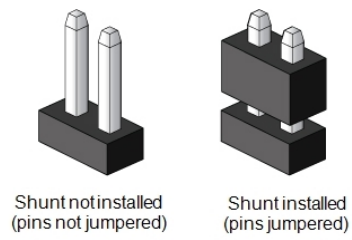


Figure 2-5: The two-pin hard jumper has two settings, jumpered and not jumpered.

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Mechanical Jumpers on the LDS6525

Table 2-7 below lists the results of each hard jumper setting.

Table 2-7: Hard Jumper Settings

Jumper	Pins	Function	Setting	Result
J36	1-2	NVMRO	Not jumpered (default)	All user-accessible LDS6525 Non-volatile memory is controlled by the state of the backplane NVMRO signal.
			Jumpered	All LDS6525 Non-volatile memory is write-protected.

Note

The state of the local J36 jumper has no affect on the state of the backplane NVMRO signal and affects no other modules in the system.

Setting Mechanical Jumpers on the LDS6525

The procedure for setting hard jumpers on the LDS6525 is as follows:

To remove or install the jumper on the module, have the following tools available:

- A pair of insulated needle-nose pliers

To set or remove the jumper on an air-cooled module:

1. Remove the module from the chassis. For instructions, see [Installing/Removing an Air-Cooled LDS6525](#) on page 53.
2. Using the insulated needle-nose pliers, install or remove the two-pin jumper.
3. Install the module into the chassis.

Non-volatile Memory Devices

The LDS6525 has both modifiable and non-modifiable Non-volatile memory devices.

Modifiable Non-volatile Memory

The following tables provide information on the *modifiable* Non-volatile memory devices on the LDS6525.

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Table 2-8: Non-volatile Memory Information—BIOS Flash

Component	BIOS Flash
Quantity on Module	Two
Purpose	Code/data storage
Type	Flash
Size	16 MB per device
How is it modified?	In-circuit by the processor, through the P-FPGA
Clearing Process	1. Ensure that the module is write-enabled 2. Erase flash or program flash with scrubbing date values

Table 2-9: Non-volatile Memory Information—SmartFusion eNVM

Component	SmartFusion eNVM
Quantity on Module	One
Purpose	Embedded flash
Type	Flash
Size	512 KB
How is it modified?	SmartFusion software
Clearing Process	Erase using SmartFusion software

Table 2-10: Non-volatile Memory Information—IPMC EEPROM

Component	IPMC EEPROM
Quantity on Module	One
Purpose	Store module status information, e.g., sensor records
Type	EEPROM
Size	256Kb
How is it modified?	In-circuit via the IPMC in the M-FPGA
Clearing Process	1. Ensure that the module is write-enabled 2. Erase EEPROM or program EEPROM with scrubbing date values by sending messages to the IPMC

Table 2-11: Non-volatile Memory Information—SATA NANDrive

Component	SATA NANDrive (flash)
Quantity on Module	One
Purpose	Code/data storage

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Type	Flash
Size	8 GB
How is it modified?	In-circuit by the processor, through the P-FPGA
Clearing Process	1. Ensure that the module is write-enabled 2. Erase flash or program flash with scrubbing data values

Table 2-12: Non-volatile Memory Information—TPM

Component	TPM
Quantity on Module	One
Purpose	Trusted Platform Management
Type	EEPROM
Size	3072 bytes
How is it modified?	In-circuit through the processor and the P-FPGA
Clearing Process	1. Ensure that the module is write-enabled 2. Erase EEPROM and/or program EEPROM with scrubbing data values

Table 2-13: Non-volatile Memory Information—MRAM

Component	MRAM
Quantity on Module	One
Purpose	Data storage
Type	Magnetoresistive RAM
Size	512KB
How is it modified?	In-circuit via the processor through the P-FPGA
Clearing Process	1. Ensure that the module is write enabled 2. Erase MRAM and/or program MRAM with scrubbing data values

Table 2-14: Non-volatile Memory Information—Soft Jumpers

Component	Multiplexed EEPROM
Quantity on Module	Four
Purpose	Module configuration data
Type	EEPROM
Size	6 bit

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How is it modified?	In-circuit via the IPMC in the M-FPGA
Clearing Process	1. Ensure that the module is write enabled 2. Erase EEPROM and/or program EEPROM with scrubbing data values by sending messages to the IPMC to do so

Table 2-15: Non-volatile Memory Information—FPGA Configuration Flash

Component	FPGA Configuration Flash
Quantity on Module	Two
Purpose	P-FPGA and FB-FPGA configuration information storage
Type	Flash
Size	256 MB
How is it modified?	In-circuit by P-FPGA when commanded by processor (via LPC or PCIe)
Clearing Process	1. Ensure that the module is write enabled 2. Command P-FPGA from processor to write FPGA images 2 and 3 with a value of all Zero's.

Table 2-16: Non-volatile Memory Information—P-FPGA EEPROM

Component	P-FPGA EEPROM
Quantity on Module	One
Purpose	Store module status information, e.g., sensor records
Type	EEPROM
Size	256Kb
How is it modified?	In-circuit via the processor
Clearing Process	1. Ensure that the module is write enabled 2. Erase EEPROM and/or program EEPROM with scrubbing data values by sending messages to the IPMC to do so.

Table 2-17: Non-volatile Memory Information—FB-FPGA EEPROM

Component	FB-FPGA EEPROM
Quantity on Module	One
Purpose	Store module status information, e.g., sensor records
Type	EEPROM
Size	256Kb
How is it modified?	In-circuit via the processor

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Clearing Process	1. Ensure that the module is write enabled 2. Erase EEPROM and/or program EEPROM with scrubbing data values by sending messages to the IPMC to do so
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Non-Modifiable Non-volatile Memory

The following tables provide information on the *non-modifiable* Non-volatile memory devices on the LDS6525.

Table 2-18: Non-volatile Memory Information—FRU EEPROM

Component	FRU EEPROM
Quantity on Module	One
Type	EEPROM
Purpose	Store module information
Size	256Kb

Table 2-19: Non-volatile Memory Information—Barton Hills Config EEPROM

Component	Barton Hills Config EEPROM
Quantity on Module	One
Type	EEPROM
Purpose	Store Barton Hills configuration information
Size	128Kb

Table 2-20: Non-volatile Memory Information—SmartFusion Flash (FPGA Logic)

Component	SmartFusion Device
Quantity on Module	One
Type	Flash (FPGA Logic)
Purpose	Performs power control, reset control, write enable control, sensor data management, FRU management, and other management tasks. Also provides FB-FPGA configuration information storage (in addition to the P-FPGA).
Size	2 Gb

Table 2-21: Non-volatile Memory Information—M-FPGA External Image SPI Flash

Component	M-FPGA Image Flash
Quantity on Module	One
Type	Flash

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Purpose	Store programming image for M-FPGA
Size	4MB

Module Resets

There are several ways to reset the LDS6525:

Power-on reset	When you apply power, all devices are held in reset until power is stable and the system reset is de-asserted.
Front panel switch reset	The front panel reset switch issues a hard reset to all module devices. For more information, see Reset Switch on page 12.
System controller reset	A system controller in the chassis can reset the LDS6525 through backplane connector P0, Pin B4, Signal <i>SYSRESET</i>
Soft Reset	soft resets can be executed via the Linux OS, using the <i>reboot</i> or <i>shutdown</i> command.

Chapter 3

LDS6525 Connections, Signals, and Pin Assignments

Introduction

The LDS6525 has seven VITA 46.0/VITA 65.0-compliant backplane connectors, VITA 42.0 PMC/XMC sites with connectors to support mezzanine cards, and several front-panel connectors (air-cooled modules only), which include Ethernet, USB, and serial interfaces, plus a test port for access to the P-FPGA. The pin assignments for all of these connectors are provided in the following topics.

Note

Table entries specifying "NC" indicate that the pin in that row is not connected.
 Table entries specifying "NU" indicate that the pin in that row is not used.
 Table entries specifying "GND" indicate a ground pin.
 Table entries specifying "*" indicate that the signal is asserted Active Low.

Backplane Connector Pin Assignments

Table 3-1 below lists the backplane connectors on the LDS6525 module.

Table 3-1: LDS6525 Backplane Connectors

Connector	Function
P0	Provides input voltages and system management functions
P1	Provides four 4x FB-FPGA links on the dataplane ports.
P2	Provides two x8 connections (can be configured to one x16)
P3	Provides PMC J14 user I/O connections
P4	Provides XMC J16 user I/O connections, two 1000BASE-T connections, and two 1000BASE-BX connections
P5	Provides XMC single-ended I/O
P6	Provides XMC J16 user I/O, USB, Display Port, and SATA connections

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P0 Connector Pin Assignments

The P0 connector is a 7 row, 56-pin connector used for input voltages and system management functions.

Table 3-2 below lists the pin assignments for the P0 connector.

Table 3-2: P0 Connector Pin Assignments

Pin	Row G	Row F	Row E	Row D	Row C	Row B	Row A
1	Vs1_+12V	Vs1_+12V	Vs1_+12V	No Pad	Vs2_+12V	Vs2_+12V	Vs2_+12V
2	Vs1_+12V	Vs1_+12V	Vs1_+12V	No Pad	Vs2_+12V	Vs2_+12V	Vs2_+12V
3	Vs3_+5V	Vs3_+5V	Vs3_+5V	No Pad	Vs3_+5V	Vs3_+5V	Vs3_+5V
4	SM2_IPMB- B_SCL	SM3_IPMB- B_SDA	GND	-12V_Aux	GND	SYSRESET*	NVMRO
5	GAP*	GA4*	GND	3.3V_Aux	GND	SM0_IPMB- A_SCL	SM1_IPMB- A_SDA
6	GA3*	GA2*	GND	+12V_Aux	GND	GA1*	GA0*
7	TCK	GND	TDO	TDI	GND	TMS	TRST*
8	GND	REF_CLK-	REF_CLK+	GND	AUX_CLK-	AUX_CLK+	GND

P1 Connector Pin Assignments

The P1 connector is a 7 row, 112-pin connector used primarily for the four 4x FB-FPGA links on the dataplane ports. In addition to the SRIO links, Row G has system management signals.

Table 3-3 below lists the pin assignments for the P1 connector.

Table 3-3: P1 Connector Pin Assignments

Pin	Row G	Row F	Row E	Row D	Row C	Row B	Row A
1	GDiscrete1	GND	DPA-TD0-	DPA-TD0+	GND	DPA-RD0-	DPA-RD0+
2	GND	DPA-TD1-	DPA-TD1+	GND	DPA-RD1-	DPA-RD1+	GND
3	P1-VBAT	GND	DPA-TD2-	DPA-TD2+	GND	DPA-RD2-	DPA-RD2+
4	GND	DPA-TD3-	DPA-TD3+	GND	DPA-RD3-	DPA-RD3+	GND
5	SYS_CON*	GND	DPB-TD0-	DPB-TD0+	GND	DPB-RD0-	DPB-RD0+
6	GND	DPB-TD1-	DPB-TD1+	GND	DPB-RD1-	DPB-RD1+	GND
7	NC	GND	DPB-TD2-	DPB-TD2+	GND	DPB-RD2-	DPB-RD2+
8	GND	DPB-TD3-	DPB-TD3+	GND	DPB-RD3-	DPB-RD3+	GND
9	EnvironBypass*	GND	DPC-TD0-	DPC-TD0+	GND	DPC-RD0-	DPC-RD0+

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Pin	Row G	Row F	Row E	Row D	Row C	Row B	Row A
10	GND	DPC-TD1-	DPC-TD1+	GND	DPC-RD1-	DPC-RD1+	GND
11	ChassisTest*	GND	DPC-TD2-	DPC-TD2+	GND	DPC-RD2-	DPC-RD2+
12	GND	DPC-TD3-	DPC-TD3+	GND	DPC-RD3-	DPC-RD3+	GND
13	MemoryClear*	GND	DPD-TD0-	DPD-TD0+	GND	DPD-RD0-	DPD-RD0+
14	GND	DPD-TD1-	DPD-TD1+	GND	DPD-RD1-	DPD-RD1+	GND
15	MaskableReset*	GND	DPD-TD2-	DPD-TD2+	GND	DPD-RD2-	DPD-RD2+
16	GND	DPD-TD3-	DPD-TD3+	GND	DPD-RD3-	DPD-RD3+	GND

P2 Connector Pin Assignments

The P2 Connector is a 7 row, 112-pin connector used primarily for PCIe expansion plane implemented as two x8 or one x16 PCIe link.

Table 3-4: P2 Connector Pin Assignments

Pin	Row G	Row F	Row E	Row D	Row C	Row B	Row A
1	PCIe_RST1_N	GND	EP_A0-TX-	EP_A0- TX+	GND	EP_A0-RX-	EP_A0-RX+
2	GND	EP_A1-TX-	EP_A1-TX+	GND	EP_A1-RX-	EP_A1-RX+	GND
3	PCIe_RST2_N	GND	EP_A2-TX-	EP_A2- TX+	GND	EP_A2-RX-	EP_A2-RX+
4	GND	EP_A3-TX-	EP_A3-TX+	GND	EP_A3-RX-	EP_A3-RX+	GND
5	NU_PCIE_CLK1_M	GND	EP_A4-TX-	EP_A4- TX+	GND	EP_A4-RX-	EP_A4-RX+
6	GND	EP_A5-TX-	EP_A5-TX+	GND	EP_A5-RX-	EP_A5-RX+	GND
7	NU_PCIE_CLK1_P	GND	EP_A6-TX-	EP_A6- TX+	GND	EP_A6-RX-	EP_A6-RX+
8	GND	EP_A7-TX-	EP_A7-TX+	GND	EP_A7-RX-	EP_A7-RX+	GND
9	NU_PCIE_CLK2_M	GND	EP_A8-TX-	EP_A8- TX+	GND	EP_A8-RX-	EP_A8-RX+
10	GND	EP_A9-TX-	EP_A9-TX+	GND	EP_A9-RX-	EP_A9-RX+	GND
11	NU_PCIE_CLK2_P	GND	EP_A10-TX-	EP_A10- TX+	GND	EP_A10-RX-	EP_A10-RX+
12	GND	EP_A11-TX-	EP_A11-TX+	GND	EP_A11-RX-	EP_A11-RX+	GND
13	DiagHook	GND	EP_A12-TX-	EP_A12- TX+	GND	EP_A12-RX-	EP_A12-RX+
14	GND	EP_A13-TX-	EP_A13-TX+	GND	EP_A13-RX-	EP_A13-RX+	GND
15	V5P0_USB2	GND	EP_A14-TX-	EP_A14- TX+	GND	EP_A14-RX-	EP_A14-RX+
16	GND	EP_A15-TX-	EP_A15-TX+	GND	EP_A15-RX-	EP_A15-RX+	GND

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P3 Connector Pin Assignments

The P3 Connector is a 7 row, 112-pin connector used for the PMC J14 User IO connections.

Table 3-5: P3 Connector Pin Assignments

Pin	Row G	Row F	Row E	Row D	Row C	Row B	Row A
1	RTM_I2C_SCL	GND	PMC_J24-01	PMC_J24-03	GND	PMC_J24-02	PMC_J24-04
2	GND	PMC_J24-05	PMC_J24-07	GND	PMC_J24-06	PMC_J24-08	GND
3	RTM_I2C_SCL	GND	PMC_J24-09	PMC_J24-11	GND	PMC_J24-10 XMC-J26-C01	PMC_J24-12 XMC-J26-F01
4	GND	PMC_J24-13 XMC-J26-C02	PMC_J24-15 XMC-J26-C03	GND	PMC_J24-14 XMC-J26-F02	PMC_J24-16 XMC-J26-F03	GND
5	USB2_DATA-	GND	PMC_J24-17 XMC_J26-C04	PMC_J24-19 XMC_J26-C05	GND	PMC_J24-18 XMC_J26-F04	PMC_J24-20 XMC_J26-F05
6	GND	PMC_J24-21 XMC_J26-C06	PMC_J24-23 XMC_J26-C07	GND	PMC_J24-22 XMC_J26-F06	PMC_J24-24 XMC_J26-F07	GND
7	USB2_DATA+	GND	PMC_J24-25 XMC_J26-C08	PMC_J24-27 XMC_J26-C09	GND	PMC_J24-26 XMC_J26-F08	PMC_J24-28 XMC_J26-F09
8	GND	PMC_J24-29 XMC_J26-C10	PMC_J24-31 XMC_J26-C11	GND	PMC_J24-30 XMC_J26-F10	PMC_J24-32 XMC_J26-F11	GND
9	UART_Ain_RTS or RX_D422_P	GND	PMC_J24-33 XMC_J26-C12	PMC_J24-35 XMC_J26-C13	GND	PMC_J24-34 XMC_J26-F12	PMC_J24-36 XMC_J26-F13
10	GND	PMC_J24-37 XMC_J26-C14	PMC_J24-39 XMC_J26-C15	GND	PMC_J24-38 XMC_J26-F14	PMC_J24-40 XMC_J26-F15	GND
11	UART_Ain_D or RX_D422_M	GND	PMC_J24-41 XMC_J26-C16	PMC_J24-43 XMC_J26-C17	GND	PMC_J24-42 XMC_J26-F16	PMC_J24-44 XMC_J26-F17
12	GND	PMC_J24-45 XMC_J26-C18	PMC_J24-47 XMC_J26-C19	GND	PMC_J24-46 XMC_J26-F18	PMC_J24-48 XMC_J26-F19	GND
13	UART_Aout_CTS or TX_D422_P	GND	PMC_J24-49	PMC_J24-51	GND	PMC_J24-50	PMC_J24-52
14	GND	PMC_J24-53	PMC_J24-55	GND	PMC_J24-54	PMC_J24-56	GND
15	RTM_I2C_SCL	GND	PMC_J24-01	PMC_J24-03	GND	PMC_J24-02	PMC_J24-04
16	GND	PMC_J24-05	PMC_J24-07	GND	PMC_J24-06	PMC_J24-08	GND

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P4 Connector Pin Assignments

Table 3-6: P4 Connector Pin Assignments

Pin	Row G	Row F	Row E	Row D	Row C	Row B	Row A
1	GDiscrete5	GND	XMC_J26-A05_TX1+	XMC_J26-B05_TX1-	GND	XMC_J26-D05_TX2+	XMC_J26-E05_TX2-
2	GND	XMC_J26-A07_TX3+	XMC_J26-B07_TX3-	GND	XMC_J26-D07_TX4+	XMC_J26-E07_TX4-	GND
3	GDiscrete6	GND	XMC_J26-A09_XX5+	XMC_J26-B09_XX5-	GND	XMC_J26-D09_XX6+	XMC_J26-E09_XX6-
4	GND	XMC_J26-A15_RX7+	XMC_J26-B15_RX7-	GND	XMC_J26-D15_RX8+	XMC_J26-E15_RX8-	GND
5	GDiscrete7	GND	XMC_J26-A17_RX9+	XMC_J26-B17_RX9-	GND	XMC_J26-D17_RX10+	XMC_J26-E17_RX10-
6	GND	XMC_J26-A19_XX11+	XMC_J26-B19_XX11-	GND	XMC_J26-D19_XX12+	XMC_J26-E19_XX12-	GND
7	GDiscrete8	GND	XMC_J26-A01_TX13+	XMC_J26-B01_TX13-	GND	XMC_J26-D01_TX14+	XMC_J26-E01_TX14-
8	GND	XMC_J26-A03_TX15+	XMC_J26-B03_TX15-	GND	XMC_J26-D03_TX16+	XMC_J26-E03_TX16-	GND
9	NU_ModeSel0	GND	XMC_J26-A11_RX17+	XMC_J26-B11_RX17-	GND	XMC_J26-D11_RX18+	XMC_J26-E11_RX18-
10	GND	XMC_J26-A13_RX19+	XMC_J26-B13_RX19-	GND	XMC_J26-D13_RX20+	XMC_J26-E13_RX20-	GND
11	NU_ModeSel1	GND	CP2_TD-	CP2_TD+	GND	CP2_RD-	CP2_RD+
12	GND	CP1_TD-	CP1_TD+	GND	CP1_RD-	CP1_RD+	GND
13	NU_ModeSel2	GND	ExtEnet2_DB-	ExtEnet2_DB+	GND	ExtEnet2_DA-	ExtEnet2_DA+
14	GND	ExtEnet2_DD-	ExtEnet2_DD+	GND	ExtEnet2_DC-	ExtEnet2_DC+	GND
15	NU_ModeSel3	GND	ExtEnet1_DB-	ExtEnet1_DB+	GND	ExtEnet1_DA-	ExtEnet1_DA+
16	GND	ExtEnet1_DD-	ExtEnet1_DD+	GND	ExtEnet1_DC-	ExtEnet1_DC+	GND

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P5 Connector Pin Assignments

Table 3-7: P5 Connector Pin Assignments

Pin	Row G	Row F	Row E	Row D	Row C	Row B	Row A
1	UART_Bin_D or UART_Bin_422_M	GND	NC_EP_B0- TX-	NC_EP_B0- TX+	GND	NC_EP_B0- RX-	NC_EP_B0- RX+
2	GND	EP_C1-TX-	NC_EP_B1- TX+	GND	NC_EP_B1- RX-	NC_EP_B1- RX+	GND
3	UART_Bin_RTS or UART_Bin_422_P	GND	NC_EP_B2- TX-	NC_EP_B2- TX+	GND	NC_EP_B2- RX-	NC_EP_B2- RX+
4	GND	EP_C3-TX-	NC_EP_B3- TX+	GND	NC_EP_B3- RX-	NC_EP_B3- RX+	GND
5	UART_Bout_D or UART_Bout_422_M	GND	NC_EP_B4- TX-	NC_EP_B4- TX+	GND	NC_EP_B4- RX-	NC_EP_B4- RX+
6	GND	EP_C5-TX-	NC_EP_B5- TX+	GND	NC_EP_B5- RX-	NC_EP_B5- RX+	GND
7	UART_Bout_CTS or UART_Bout_422_P	GND	NC_EP_B6- TX-	NC_EP_B6- TX+	GND	NC_EP_B6- RX-	NC_EP_B6- RX+
8	GND	EP_C7-TX-	NC_EP_B7- TX+	GND	NC_EP_B7- RX-	NC_EP_B7- RX+	GND
9	UART_Cin_D	GND	XMC_J16_C12	XMC_J16_C13	GND	XMC_J16_F12	XMC_J16_F13
10	GND	XMC_J16_C14	XMC_J16_C15	GND	XMC_J16_F14	XMC_J16_F15	GND
11	UART_Cout_D	GND	XMC_J16_C16	XMC_J16_C17	GND	XMC_J16_F16	XMC_J16_F17
12	GND	XMC_J16_C18	XMC_J16_C19	GND	XMC_J16_F18	XMC_J16_F19	GND
13	UART_Din_D	GND	USB6_TX-	USB6_TX+	GND	USB6_RX-	USB6_RX+
14	GND	USB7_TX-	USB7_TX+	GND	USB7_RX-	USB7_RX+	GND
15	UART_Dout_D	GND	USB1_DATA-	USB1_DATA+	GND	NC	NC
16	GND	SATA1_TX-	SATA1_TX+	GND	SATA1_RX-	SATA1_RX+	GND

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P6 Connector Pin Assignments

Table 3-8: P6 Connector Pin Assignments

Pin	Row G	Row F	Row E	Row D	Row C	Row B	Row A
1	GDiscrete9	GND	XMC_J16-A05_ TX1+	XMC_J16-B05_ TX1-	GND	XMC_J16-D05_ TX2+	XMC_J16-E05_ TX2-
2	GND	XMC_J16-A07_ TX3+	XMC_J16-B07_ TX3-	GND	XMC_J16-D07_ TX4+	XMC_J16-E07_ TX4-	GND
3	GDiscrete10	GND	XMC_J16-A09_ XX5+ or (FPGA_ GPIO_9)	XMC_J16-B09_ XX5- or (FPGA_ GPIO_10)	GND	XMC_J16-D09_ XX6+ or (FPGA_ GPIO_11)	XMC_J16-E09_ XX6- or (FPGA_ GPIO_12)
4	GND	XMC_J16-A15_ RX7+	XMC_J16-B15_ RX7-	GND	XMC_J16-D15_ RX8+	XMC_J16-E15_ RX8-	GND
5	GDiscrete11	GND	XMC_J16-A17_ RX9+	XMC_J16-B17_ RX9-	GND	XMC_J16-D17_ RX10+	XMC_J16-E17_ RX10-
6	GND	XMC_J16-A19_ XX11+ or (FPGA_GPIO_ 13)	XMC_J16-B19_ XX11- or (FPGA_GPIO_ 14)	GND	XMC_J16-D19_ XX12+ or (FPGA_GPIO_ 15)	XMC_J16-E19_ XX12- or (FPGA_GPIO_ 16)	GND
7	GDiscrete12	GND	XMC_J16-A01_ TX13+	XMC_J16-B01_ TX13-	GND	XMC_J16-D01_ TX14+	XMC_J16-E01_ TX14-
8	GND	XMC_J16-A03_ TX15+	XMC_J16-B03_ TX15-	GND	XMC_J16-D03_ TX16+	XMC_J16-E03_ TX16-	GND
9	V5P0_DP2	GND	XMC_J16-A11_ RX17+	XMC_J16-B11_ RX17-	GND	XMC_J16-D11_ RX18+	XMC_J16-E11_ RX18-
10	GND	XMC_J16-A13_ RX19+	XMC_J16-B13_ RX19-	GND	XMC_J16-D13_ RX20+	XMC_J16-E13_ RX20-	GND
11	V5P0_DP1	GND	DP1_LANE0_P	DP1_LANE0_M	GND	DP1_LANE1_P	DP1_LANE1_M
12	GND	DP1_LANE2_P	DP1_LANE2_M	GND	DP1_LANE3_P	DP1_LANE3_M	GND
13	V5P0_USB1	GND	USB1_Data-	USB1_DATA+	GND	NC	NC
14	GND	USB10_TX-	USB10_TX+	GND	USB10_RX-	USB10_RX+	GND
15	DVI1_HPD	GND	SATA1_TX-	SATA1_TX+	GND	SATA1_RX-	SATA1_RX+
16	GND	SATA2_TX-	SATA2_TX+	GND	SATA2_RX-	SATA2_RX+	GND

PMC/XMC Connector Pin Assignments

Table 3-9 below lists the mezzanine site connectors on the LDS6525 module.

Table 3-9: LDS6525 Mezzanine Site Connectors

Connector	Function
J11, J12, J13, J14	Site 1 PMC connectors
J15, J16	Site 1 XMC connectors
J25, J26	Site 2 XMC connectors

J11 Connector Pin Assignments

Pin	Signal	Signal	Pin
1	TCK_MOD	VM12P_AUX	2
3	GND	PCI_2_INTA_N	4
5	PCI_2_INTB_N	PCI_2_INTC_N	6
7	PCI_2_BUSMODE1_N	V5P0	8
9	PCI_2_INTD_N	NC	10
11	GND	NC	12
13	PCI_2_PCI_CLK	GND	14
15	GND	PCI_2_GNT0_N	16
17	PCI_2_REQ0_N	V5P0	18
19	V3P3_XMC	PCI_2_AD31	20
21	PCI_2_AD28	PCI_2_AD27	22
23	PCI_2_AD25	GND	24
25	GND	PCI_2_CBE3	26
27	PCI_2_AD22	PCI_2_AD21	28
29	PCI_2_AD19	V5P0	30
31	V3P3_XMC	PCI_2_AD17	32
33	PCI_2_FRAME_N	GND	34
35	GND	PCI_2_IRDY_N	36
37	PCI_2_DEVSEL_N	V5P0	38
39	PCI_2_PCIXCAP	PCI_2_LOCK_N	40

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Pin	Signal	Signal	Pin
41	NC	NC	42
43	PCI_2_PAR	GND	44
45	V3P3_XMC	PCI_2_AD15	46
47	PCI_2_AD12	PCI_2_AD11	48
49	PCI_2_AD9	V5P0	50
51	GND	PCI_2_CBE0	52
53	PCI_2_AD6	PCI_2_AD5	54
55	PCI_2_AD4	GND	56
57	V3P3_XMC	PCI_2_AD3	58
59	PCI_2_AD2	PCI_2_AD1	60
61	PCI_2_AD0	V5P0	62
63	GND	PCI_2_REQ64_N	64

J12 Connector Pin Assignments

Pin	Signal	Signal	Pin
1	V12P_AUX	TRSTN_MOD	2
3	TMS_MOD	TDO_XMC2	4
5	TDI_XMC2	GND	6
7	GND	NC	8
9	NC	NC	10
11	PCI_2_MODE2_PU_N	V3P3_XMC	12
13	PCI_2_RST_N	PCI_2_MODE3_PD_N	14
15	V3P3_XMC	PCI_2_MODE4_PD_N	16
17	PCI_2_PME_N	GND	18
19	PCI_2_AD30	PCI_2_AD29	20
21	GND	PCI_2_AD26	22
23	PCI_2_AD24	V3P3_XMC	24
25	PCI_2_IDSEL	PCI_2_AD23	26
27	V3P3_XMC	PCI_2_AD20	28
29	PCI_2_AD18	GND	30

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Pin	Signal	Signal	Pin
31	PCI_2_AD16	PCI_2_CBE2	32
33	GND	NC	34
35	PCI_2_TRDY_N	V3P3_XMC	36
37	GND	PCI_2_STOP_N	38
39	PCI_2_PERR_N	GND	40
41	V3P3_XMC	PCI_2_SERR_N	42
43	PCI_2_CBE1	GND	44
45	PCI_2_AD14	PCI_2_AD13	46
47	PCI_2_M66EN	PCI_2_AD10	48
49	PCI_2_AD8	V3P3_XMC	50
51	PCI_2_AD7	NC	52
53	V3P3_XMC	NC	54
55	NC	GND	56
57	NC	NC	58
59	GND	NC	60
61	PCI_2_ACK64_N	V3P3_XMC	62
63	GND	NC	64

J13 Connector Pin Assignments

Pin	Signal	Signal	Pin
1	NC	GND	2
3	GND	PCI_2_CBE7_N	4
5	PCI_2_CBE6_N	PCI_2_CBE5_N	6
7	PCI_2_CBE4_N	GND	8
9	V3P3_XMC	PCI_2_PAR64	10
11	PCI_2_AD63	PCI_2_AD62	12
13	PCI_2_AD61	GND	14
15	GND	PCI_2_AD60	16
17	PCI_2_AD59	PCI_2_AD58	18
19	PCI_2_AD57	GND	20
21	V3P3_XMC	PCI_2_AD56	22

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Pin	Signal	Signal	Pin
23	PCI_2_AD55	PCI_2_AD54	24
25	PCI_2_AD53	GND	26
27	GND	PCI_2_AD52	28
29	PCI_2_AD51	PCI_2_AD50	30
31	PCI_2_AD49	GND	32
33	GND	PCI_2_AD48	34
35	PCI_2_AD47	PCI_2_AD46	36
37	PCI_2_AD45	GND	38
39	V3P3_XMC	PCI_2_AD44	40
41	PCI_2_AD43	PCI_2_AD42	42
43	PCI_2_AD41	GND	44
45	GND	PCI_2_AD40	46
47	PCI_2_AD39	PCI_2_AD38	48
49	PCI_2_AD37	GND	50
51	GND	PCI_2_AD36	52
53	PCI_2_AD35	PCI_2_AD34	54
55	PCI_2_AD33	GND	56
57	V3P3_XMC	PCI_2_AD32	58
59	NC	NC	60
61	NC	GND	62
63	GND	NC	64

J14 Connector Pin Assignments

Pin	P3	Pin	P3	Pin	P3	Pin	P3
1	E1	2	B1	33	E9	34	B9
3	D1	4	A1	35	D9	36	A9
5	F2	6	C2	37	F10	38	C10
7	E2	8	B2	39	E10	40	B10
9	E3	10	B3	41	E11	42	B11

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Pin	P3	Pin	P3	Pin	P3	Pin	P3
11	D3	12	A3	43	D11	44	A11
13	F4	14	C4	45	F12	46	C12
15	E4	16	B4	47	E12	48	B12
17	E5	18	B5	49	E13	50	B13
19	D5	20	A5	51	D13	52	A13
21	F6	22	C6	53	F14	54	C14
23	E6	24	B6	55	E14	56	B14
25	E7	26	B7	57	E15	58	B15
27	D7	28	A7	59	D15	60	A15
29	F8	30	C8	61	F16	62	C16
31	E8	32	B8	63	E16	64	B16

J15 Connector Pin Assignments

Table 3-10: J15

Pin	A	B	C	D	E	F
1	XMC1_RX_0_P	XMC1_RX_0_M	V3P3	XMC1_RX_1_P	XMC1_RX_1_M	V12P0
2	GND	GND	TRSTN_MOD	GND	GND	XMCJ15_RSTIN_N
3	XMC1_RX_2_P	XMC1_RX_2_M	V3P3	XMC1_RX_3_P	XMC1_RX_3_M	V12P0
4	GND	GND	TCK_MOD	GND	GND	XMCJ15_RSTOUT_N
5	XMC1_RX_4_P	XMC1_RX_4_M	V3P3	XMC1_RX_5_P	XMC1_RX_5_M	V12P0
6	GND	GND	TMS_MOD	GND	GND	V12P_AUX
7	XMC1_RX_6_P	XMC1_RX_6_M	V3P3	XMC1_RX_7_P	XMC1_RX_7_M	V12P0
8	GND	GND	TDI_XMC1	GND	GND	VM12P_AUX
9	PCIE_RX_P	PCIE_RX_M	NC_UD_J15_C9	PCIE_TX_P	PCIE_TX_M	V12P0
10	GND	GND	TDO_XMC1	GND	GND	XMC1_GA0
11	XMC1_TX_0_AC_P	XMC1_TX_0_AC_M	XMC_BIST_N	XMC1_TX_1_AC_P	XMC1_TX_1_AC_M	V12P0
12	GND	GND	XMC1_GA1	GND	GND	XMCJ15_PRESENT_N
13	XMC1_TX_2_AC_P	XMC1_TX_2_AC_M	V3P3	XMC1_TX_3_AC_P	XMC1_TX_3_AC_M	V12P0
14	GND	GND	XMC1_GA2	GND	GND	IPMI_SDA1
15	XMC1_TX_4_AC_P	XMC1_TX_4_AC_M	NC	XMC1_TX_5_AC_P	XMC1_TX_5_AC_M	V12P0

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Pin	A	B	C	D	E	F
16	GND	GND	XMC_WP	GND	GND	IPMI_SCL1
17	XMC1_TX_6_AC_P	XMC1_TX_6_AC_M	NC	XMC1_TX_7_AC_P	XMC1_TX_7_AC_M	NC
18	GND	GND	NC	GND	GND	NC
19	XMC_100M_REFCLK_P	XMC_100M_REFCLK_M	NC	NC_XMC1_WAKE_N	NC_XMC1_ROOT0_N	NC

J16 Connector Pin Assignments

Table 3-11: J16

Pin	A		B		C		D		E		F	
1	P6-E7	TX13+	P6-D7	TX13-	NC	NC	P6-B7	TX14+	P6-A7	TX14-	NC	
2	GND	GND	GND	GND	NC	NC	GND	GND	GND	GND	NC	
3	P6-F8	TX15+	P6-E8	TX15-	NC	NC	P6-C8	TX16+	P6-B8	TX16-	NC	
4	GND	GND	GND	GND	NC	NC	GND	GND	GND	GND	NC	
5	P6-E1	TX1+	P6-D1	TX1-	NC	NC	P6-B1	TX2+	P6-A1	TX2-	NC	
6	GND	GND	GND	GND	NC	NC	GND	GND	GND	GND	NC	
7	P6-F2	TX3+	P6-E2	TX3-	NC	NC	P6-C2	TX4+	P6-B2	TX4-	NC	
8	GND	GND	GND	GND	NC	NC	GND	GND	GND	GND	NC	
9	P6-E3	XX5+	P6-D3	XX5-	NC	NC	P6-B3	XX6+	P6-A3	XX6-	NC	
10	GND	GND	GND	GND	NC	NC	GND	GND	GND	GND	NC	
11	P6-E9	RX17+	P6-D9	RX17-	NC	NC	P6-B9	RX18+	P6-A9	RX18-	NC	
12	GND	GND	GND	GND	P5-E9	SE33	GND	GND	GND	GND	P5-B9	SE34
13	P6-F10	RX19+	P6-E10	RX19-	P5-D9	SE35	P6-C10	RX20+	P6-B10	RX20-	P5-A9	SE36
14	GND	GND	GND	GND	P5-F10	SE37	GND	GND	GND	GND	P5-C10	SE38
15	P6-F4	RX7+	P6-E4	RX7-	P5-E10	SE39	P6-C4	RX8+	P6-B4	RX8-	P5-B10	SE40
16	GND	GND	GND	GND	P5-E11	SE41	GND	GND	GND	GND	P5-B11	SE42
17	P6-E5	RX9+	P6-D5	RX9-	P5-D11	SE43	P6-B5	RX10+	P6-A5	RX10-	P5-A11	SE44
18	GND	GND	GND	GND	P5-F12	SE45	GND	GND	GND	GND	P5-C12	SE46
19	P6-F6	XX11+	P6-E6	XX11-	P5-E12	SE47	P6-C6	XX12+	P6-B6	XX12-	P5-B12	SE48

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J25 Connector Pin Assignments

Pin	A	B	C	D	E	F
1	XMC2_RX_0_P	XMC2_RX_0_M	V3P3_XMC	XMC2_RX_1_P	XMC2_RX_1_M	V12P0
2	GND	GND	TRSTN_MOD	GND	GND	XMCJ25_RSTIN_N
3	XMC2_RX_2_P	XMC2_RX_2_M	V3P3_XMC	XMC2_RX_3_P	XMC2_RX_3_M	V12P0
4	GND	GND	TCK_MOD	GND	GND	XMCJ25_RSTOUT_N
5	XMC2_RX_4_P	XMC2_RX_4_M	V3P3_XMC	XMC2_RX_5_P	XMC2_RX_5_M	V12P0
6	GND	GND	TMS_MOD	GND	GND	V12P_AUX
7	XMC2_RX_6_P	XMC2_RX_6_M	V3P3_XMC	XMC2_RX_7_P	XMC2_RX_7_M	V12P0
8	GND	GND	TDI_XMC1	GND	GND	VM12P_AUX
9	PCIE_RX_P	PCIE_RX_M	NC_UD_J25_C9	PCIE_TX_P	PCIE_TX_M	V12P0
10	GND	GND	TDO_XMC1	GND	GND	XMC2_GA0
11	XMC2_TX_0_AC_P	XMC2_TX_0_AC_M	XMC_BIST_N	XMC2_TX_1_AC_P	XMC2_TX_1_AC_M	V12P0
12	GND	GND	XMC2_GA1	GND	GND	XMCJ25_PRESENT_N
13	XMC2_TX_2_AC_P	XMC2_TX_2_AC_M	V3P3_XMC	XMC2_TX_3_AC_P	XMC2_TX_3_AC_M	V12P0
14	GND	GND	XMC2_GA2	GND	GND	IPMI_SDA1
15	XMC2_TX_4_AC_P	XMC2_TX_4_AC_M	NC	XMC2_TX_5_AC_P	XMC2_TX_5_AC_M	V12P0
16	GND	GND	XMC_WP	GND	GND	IPMI_SCL1
17	XMC2_TX_6_AC_P	XMC2_TX_6_AC_M	NC	XMC2_TX_7_AC_P	XMC2_TX_7_AC_M	NC
18	GND	GND	NC	GND	GND	NC
19	XMC_100M_REFCLK_P	XMC_100M_REFCLK_M	NC	NC_XMC2_WAKE_N	NC_XMC2_ROOT0_N	NC

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J26 Connector Pin Assignments

Pin	A		B		C		D		E		F	
1	P4-E7	TX13+	P4-D7	TX13-	NC	SE10	P4-B7	TX14+	P4-A7	TX14-	NC	SE12
2	GND	GND	GND	GND	P3-F4	SE13	GND	GND	GND	GND	P3-C4	SE14
3	P4-F8	TX15+	P4-E8	TX15-	P3-E4	SE15	P4-C8	TX16+	P4-B8	TX16-	P3-B4	SE16
4	GND	GND	GND	GND	P3-E5	SE17	GND	GND	GND	GND	P3-B5	SE18
5	P4-E1	TX1+	P4-D1	TX1-	P3-D5	SE19	P4-B1	TX2+	P4-A1	TX2-	P3-A5	SE20
6	GND	GND	GND	GND	P3-F6	SE21	GND	GND	GND	GND	P3-C6	SE22
7	P4-F2	TX3+	P4-E2	TX3-	P3-E6	SE23	P4-C2	TX4+	P4-B2	TX4-	P3-B6	SE24
8	GND	GND	GND	GND	P3-E7	SE25	GND	GND	GND	GND	P3-B7	SE26
9	P4-E3	XX5+	P4-D3	XX5-	P3-D7	SE27	P4-B3	XX6+	P4-A3	XX6-	P3-A7	SE28
10	GND	GND	GND	GND	P3-F8	SE29	GND	GND	GND	GND	P3-C8	SE30
11	P4-E9	RX17+	P4-D9	RX17-	P3-E8	SE31	P4-B9	RX18+	P4-A9	RX18-	P3-B8	SE32
12	GND	GND	GND	GND	P3-E9	SE33	GND	GND	GND	GND	P3-B9	SE34
13	P4-F10	RX19+	P4-E10	RX19-	P3-D9	SE35	P4-C10	RX20+	P4-B10	RX20-	P3-A9	SE36
14	GND	GND	GND	GND	P3-F10	SE37	GND	GND	GND	GND	P3-C10	SE38
15	P4-F4	RX7+	P4-E4	RX7-	P3-E10	SE39	P4-C4	RX8+	P4-B4	RX8-	P3-B10	SE40
16	GND	GND	GND	GND	P3-E11	SE41	GND	GND	GND	GND	P3-B11	SE42
17	P4-E5	RX9+	P4-D5	RX9-	P3-D11	SE43	P4-B5	RX10+	P4-A5	RX10-	P3-A11	SE44
18	GND	GND	GND	GND	P3-F12	SE45	GND	GND	GND	GND	P3-C12	SE46
19	P4-F6	XX11+	P4-E6	XX11-	P3-E12	SE47	P4-C6	XX12+	P4-B6	XX12-	P3-B12	SE48

J14 to P3 Interconnect

J14	P3	J14	P3	J14	P3	J14	P3
1	E1	2	B1	33	E9	34	B9
3	D1	4	A1	35	D9	36	A9
5	F2	6	C2	37	F10	38	C10
7	E2	8	B2	39	E10	40	B10
9	E3	10	B3	41	E11	42	B11
11	D3	12	A3	43	D11	44	A11
13	F4	14	C4	45	F12	46	C12

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J14	P3	J14	P3	J14	P3	J14	P3
15	E4	16	B4	47	E12	48	B12
17	E5	18	B5	49	E13	50	B13
19	D5	20	A5	51	D13	52	A13
21	F6	22	C6	53	F14	54	C14
23	E6	24	B6	55	E14	56	B14
25	E7	26	B7	57	E15	58	B15
27	D7	28	A7	59	D15	60	A15
29	F8	30	C8	61	F16	62	C16
31	E8	32	B8	63	E16	64	B16

Front Panel Connector Pin Assignments

The air-cooled LDS6525 provides front panel connections for USB, Gigabit Ethernet, and serial communication, plus a test port for debugging the P-FPGA.

Micro DB-9 Serial Connector Pin Assignments

The front panel micro DB-9 connector provides a serial RS-232 interface to the onboard CRB through the P-FPGA UART-B. This connector provides an interface to the Intel Core i7, and another interface to the system management H8. These interfaces signal at RS-232 levels. A secondary front panel UART connection can be configured to connect to the on-board IPMC (M-FPGA2 UART-0) or a processor UART interface (P-FPGA UART-C). For more information, see UART Port Configuration on page 72.

Table 3-12 below lists the primary (FP-UART-A) and secondary (FP-UART-B) serial UART interfaces to components on the LDS6525.

Table 3-12: Micro DB-9 Serial Connector Pins

Pin Number	Signal Assignment
1	FP_UARTB_RS232_TX_F
2	FP_UARTA_RS232_TX_F
3	FP_UARTA_RS232_RX_F
4	FP_UARTB_RS232_RX_F
5	GND
6	FP_FAILSAFE_N
7	FP_UARTA_CTS_RTS

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Pin Number	Signal Assignment
8	FP_UARTA_CTS_RTS
9	FP_MS

USB Connector Pin Assignments

The LDS6525 front panel USB connector supports USB 2.0 and connects directly to port 5 of the PCH USB controller (EHCI 1). [Table 3-13](#) below lists the USB connector pin assignments.

Table 3-13: USB Connector Pin Assignments

Pin	Signal Assignment	Description
U3	USB2P5_P	USB positive differential used for address/data
U2	USB2P5_M	USB negative differential used for address/data
U1	V5P0	Power
U4	GND	Ground

RJ-45 Ethernet Connector Pin Assignments

The front panel RJ-45 Ethernet connector provides one 10/100/1000BASE-T Ethernet port. Signaling across the pins depends on the Ethernet physical layer interface being used. [Table 3-14](#) below lists the pin assignments for the front panel RJ45 connector.

Table 3-14: RJ-45 Ethernet Connector Pin Assignments

Pin	10/100BASE-T	1000BASE-T
1	TX_P	BI_DA_P
2	TX_M	BI_DA_M
3	RX_P	BI_DB_P
4	No Connection	BI_DC_P
5	No Connection	BI_DC_M
6	RX_M	BI_DB_M
7	No Connection	BI_DD_P
8	No Connection	BI_DD_M

FPGA Test Port Connector Pin Assignments

The front panel 16-bit single-ended I/O test port is a 24-pin header provided as an interface to the P-FPGA.

Table 3-15 below lists the pin assignments for the front panel FPGA test port.

Table 3-15: FPGA Test Port Connector Pin Assignments

Pin	Signal Name	Pin	Signal Name
1	TEST_PORT_00	2	+1.8V
3	TEST_PORT_01	4	TEST_PORT_02
5	GND	6	TEST_PORT_03
7	TEST_PORT_04	8	GND
9	TEST_PORT_05	10	TEST_PORT_06
11	GND	12	TEST_PORT_07
13	TEST_PORT_08	14	GND
15	TEST_PORT_09	16	TEST_PORT_10
17	GND	18	TEST_PORT_11
19	TEST_PORT_12	20	GND
21	TEST_PORT_13	22	TEST_PORT_14
23	+1.8V	24	TEST_PORT_15

Chapter 4

LDS6525 Specifications

Environmental Specifications

The LDS6525 environmental specifications are based on MIL-STD-810F.

Air-cooled Environmental Specifications

The air-cooled LDS6525 is intended for use in laboratory environments only. When inserted in a VPX air-cooled system, the air-cooled LDS6525 meets the parameters listed in [Table 4-1](#) below.

Table 4-1: Air-Cooled LDS6525 Environmental Specifications

Parameter	Value	Standard
Operating temperature	-25 to +55 °C (-13to +131 °F) (inlet air temperature)	MIL-STD-810F
Storage temperature	-55 to +85 °C (-67 to +185 °F)	MIL-STD-810F, Method 501.4, 502.4
Humidity	5% to 95% non-condensing	MIL-STD-810F, Method 507.4, notice 3 (Aggravated)
Vibration, Random	0.04 G ² /Hz, 20 to 2,000 Hz, 1 hr/axis	MIL-STD-810F, Method 514.5
Vibration, Shock	Z-axis: 50 Gs, X/Yaxes: 80 Gs 11-msec 1/2 sine pulse, 3 positive, 3 negative	MIL-STD-810F, Method 516.5, Procedure 1
Altitude, operating ¹	0 to 9,144 meters (0 to 30,000 feet)	
Altitude, storage	0 to 15,240 meters (0 to 50,000 feet)	

The chassis in which the LDS6525 is configured must supply 15 CFM for the slot occupied by the LDS6525, with a .5-inch water pressure drop across the module.

¹Operating altitudes are reached by increasing airflow or decreasing inlet air temperature.

Electrical Specifications

The LDS6525 has the following power requirements.

Power Entry

To power the chassis manager-related components, the LDS6525 takes 3.3V AUX power from the backplane. To generate the payload 3.3V, 2.5V and 1.2V rails, the LDS6525 takes 12V plane power from the backplane.

Input Power Protection

The 12V input has an electronic circuit breaker (ESB) that is current-limited to 22 Amps. When you insert the module, or turn on the main power supply voltages, the ESB limits the in-rush current from the backplane voltages. In addition, the ESB protects the module against over-current or short-circuit conditions.

Power Consumption

Total power dissipation for the LDS6525, including power conversion inefficiencies, is 63 Watts maximum from the 12V input.

Physical Characteristics

The LDS6525 has the following physical characteristics.

Dimensions

The air-cooled LDS6525 is a 1.00-inch-pitch plug-in unit that meets the dimensional specifications put forth in VITA 48.1, *Mechanical Specifications for Micro-computers Using REDI Air Cooling Applied to VITA 46*.

Weight

A fully assembled air-cooled LDS6525 weighs 993.3 gms (2.19 lbs).

Product MTBF Report

The following is the Mercury Product MTBF Report for the LDS6525.

Board Name: *LDS6525 6U VPX Air Cooled Single Haswell Poet 8G DDR3 AT 1600MHZ REV 3 Complete PCA ASSY Board Code:MXU.*

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MTBF Prediction Description

The Mean Time Between Failures (MTBF) predictions are an estimate of the basic reliability for a field replaceable unit (FRU). Many Mercury products have MTBF predictions using two typical industry methods: Telcordia® SR-332-1 and MIL-Handbook-217F1.

The Telcordia MTBF is based on the predicted parts count failure rate from Telcordia SR-332, Issue 1, Method 1, and Case 1 for Comfort Controlled conditions.

- The Quality Factor multiplier is 1.0.
- The assumed conditions are 40°C ambient air temperature and 50% electrical stress.
- At Mercury, the SR-332 Issue 1 MTBF is identical to the SR-332 Issue 2 MTBF. The SR-332 (1) is a standard failure rate prediction method, but MTBF realism must be established using field data. The SR-332 failure rate is measured in Failures In Time (FIT), where one FIT is one failure per billion hours (FPBH). The SR-332 MTBF is equal to $1,000,000,000/\text{SR-332 failure rate in FITs}$.

The MIL-Handbook MTBF is based on the MIL-Handbook-217F Notice 1 Parts Count prediction for failure rate at Ground Benign (GB).

- The Quality Factor multiplier is 1.0.
- The 217F1 Parts Count GB assumptions are 30°C ambient air and 50% electrical stress.
- The 217F1 failure rate is measured in failures per million hours (FPMH). The 217F1 MTBF is equal to $1,000,000/217F1 \text{ failure rate in FPMH}$.
- MIL-Handbook-217F1 is a standard failure rate prediction method, but MTBF realism must be established using field data. For more information, see [MTBF Realism](#) on page 99..

Prediction Methods and Results

The following table provides methods, results, and scales for the LDS6525 MTBF data provided in this document.

Table 4-2: Prediction Methods and Results

Method	Result	Scale
Telcordia SR-332(1) Parts Count, Comfort Controlled, 40C, 50% stress, Quality Multiplier =1	9032.4	SR-332 FR in failures per billion hours
	111	SR-332 MTBF in Khrs
MIL-Handbook-217F(1) Parts Count, Ground Benign, 30C, 50% stress, Quality Multiplier =1	13.26827	217F1 FR in failures per million hours
	75	217F1 MTBF in Khrs

Thermal and Environmental Variations

Thermal and Environmental variations are based on QuARTPRO V2.0 Release 1 Build 73 for MIL-Handbook-217F1 (Quanterion.com).

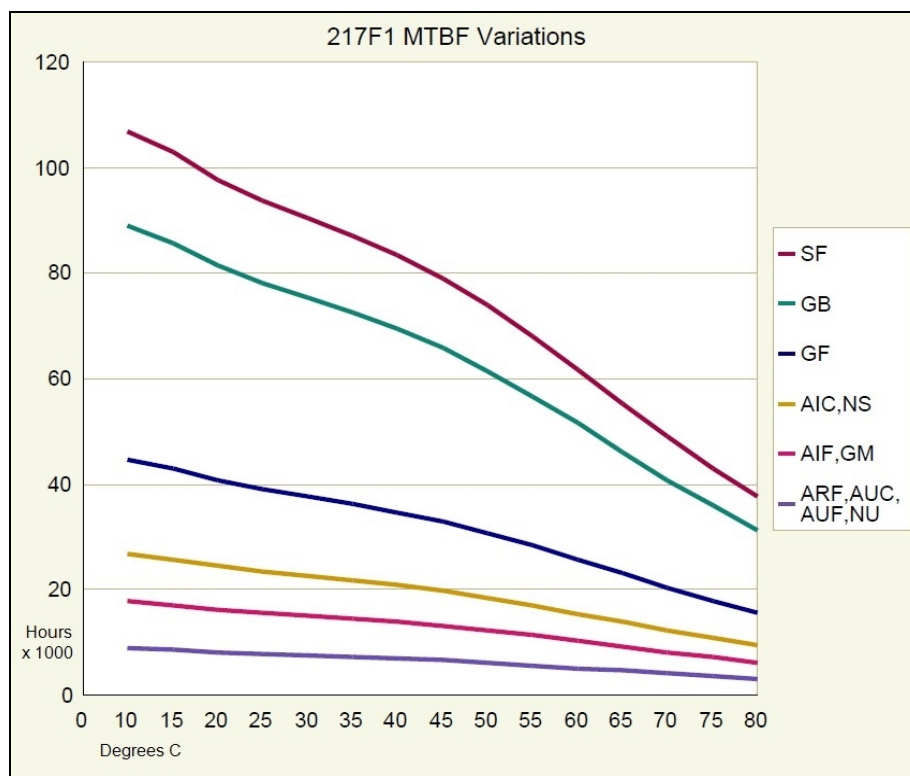


Figure 4-1: LDS6525 217F1 Variations

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Environmental Conditions

O	217F MTBF Origin	AIC	Airborne Inhabited Cargo	AIF	Airborne Inhabited Fighter	ARF	Airborne Rotary Flight
AUC	Airborne Uninhabited Cargo	AUF	Airborne Uninhabited Fighter	GB	Ground Benign	GF	Ground Fixed
GM	Ground Mobile	NS	Naval Sheltered	NU	Naval Unsheltered	SF	Space Flight

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Chapter 5

LDS6525 Installation

 CAUTION	This product contains electrostatic-sensitive components. Make sure to follow standard ESD precautions when installing, removing, or otherwise handling this product. For more information, see ESD Precautions on page 97.
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Do not attempt to install the LDS6525 in an unsupported chassis. For information on the supported chassis and other configuration information, see [System Configurations](#) on page 1.

 CAUTION	The LDS6525 is not hot-swappable. You must remove all power before removing the module from, or inserting the module into a chassis.
---	--

Installing/Removing an Air-Cooled LDS6525

After you install the LDS6525 in your chassis, you must make connections to facilitate configuration. For information on making module connections, see [Making Connections](#) on page 56. For information on configuring the LDS6525, see the *Ensemble 6000 Series Configuration Guide*.

Tool Requirements

To install or remove an LDS6525, have the following tools available:

- #1 Phillips screwdriver
- Wrist grounding strap

In addition to these tools, to support system airflow and cooling requirements, you will need slot blockers to fill empty slots in the chassis. To install or remove a slot blocker, you might need a 3/16-inch slotted screwdriver.

Installing an Air-Cooled LDS6525



This product contains electrostatic-sensitive components. Make sure to follow standard ESD precautions when installing, removing, or otherwise handling this product. For more information, see [ESD Precautions](#) on page 97.

To install the LDS6525 in a chassis:

1. Turn off main and auxiliary power to the chassis.
2. Put on the wrist grounding strap.
Make sure to observe the recommendations in [ESD Precautions](#) on page 97.
3. Remove the module from its ESD packaging.
4. Open the ejector handles:
 - a. Unlock each handle by pressing the lock button.
 - b. Rotate the ejector handles outward.

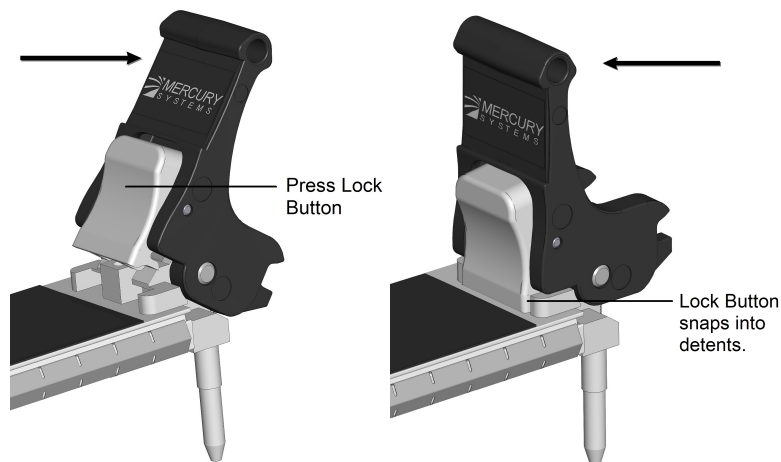


Figure 5-1: Ejector handles on a Mercury air-cooled module.

5. Holding the module with the primary side toward the right, align the module with the slot guides on the chassis.
Note: The slot must be configured for slot profile VITA48. For more information, see [Compliance](#) on page 7.
6. Applying even pressure to the upper and lower handles, insert the module into the slot until you feel the module stop against the backplane.

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 CAUTION	Do not force the module into the slot. Only gentle force should be required to insert the module. If you are applying more pressure: 1. Remove the module from the chassis and make sure that no connector pins are bent. If any pin is bent, do not install the module and do not attempt to fix the pin yourself. For assistance, contact Mercury Global Support Services. 2. Ensure that the module is properly aligned in the slot guides. 3. Try inserting the module again.
---	--

7. Rotate the ejector handles to the locked position.
8. To secure the module to the chassis, use the #1 Phillips screwdriver to tighten the two mounting screws.

The screws are located behind the ejector handles, as shown below.

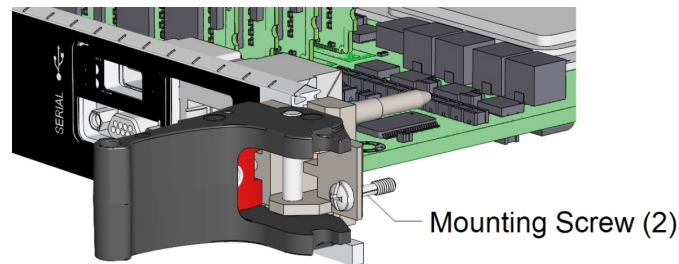


Figure 5-2: Mounting screw locations on the air-cooled LDS6525.

Removing an Air-Cooled LDS6525

To remove the LDS6525 from a chassis:

1. Turn off main and auxiliary power to the chassis.
2. Put on a wrist grounding strap.
Make sure to observe the recommendations in [ESD Precautions](#) on page 97.
3. Using the #1 Phillips screwdriver, loosen the mounting screws behind the ejector handles. See [Figure 5-2](#) above.
4. Open the ejector handles (see on page 53):
 - a. Unlock each handle by pressing the lock button.
 - b. Rotate the ejector handles outward.
5. Pulling evenly on the upper and lower ejector handles, slide the module out of

the chassis slot.

6. Place the module on a stable, dry surface, preferably inside ESD packaging.
7. Install a slot blocker in the empty chassis slot.

Making Connections

The air-cooled LDS6525 has Ethernet and serial ports on the front panel, so you can connect directly to those ports using the appropriate cables.

Establishing a Serial Connection

The default serial communication settings for the LDS6525 modules are:

- Baud rate: 115200
- Data bits: 8
- Parity: none
- Stop bits: 1
- Flow control: none

To change the serial port settings:

1. On the host, to set the required speed, use the controls provided by your serial port program (Minicom, HyperTerminal, and so on).
2. On the module, use the BIOS management tools. For more information, see [BIOS Management](#) on page 1.

To establish a serial connection from your development machine to an air-cooled LDS6525 module:

1. Turn off main and auxiliary power to the chassis.
2. Using a 9-pin D-SUB to Micro-D cable (Mercury part number 911-23011), connect the target module serial port to the development machine.

To open a console session, use a terminal emulator (e.g., Minicom or HyperTerminal), or telnet to a port server (if a serial-to-Ethernet port controller is used).

[Figure 1-1](#) on page 1 shows the location of the serial port on the air-cooled LDS6525.

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Establishing an Ethernet Connection

To establish an Ethernet connection from the development machine to an air-cooled LDS6525 module:

- Using a standard Ethernet cable, connect the LDS6525 front panel Ethernet port to a port on your network.

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Chapter 6

System Management

Overview

Using Mercury-provided system management functions, you can access LDS6525 FRU information, sensor data, and soft jumpers.

System management events are generated whenever a sensor threshold is exceeded. Once generated, these events are stored in the local system event log. They are also sent to the remote Chassis Management Controller (ChMC) and stored in the Chassis Manager's system event log.

In addition to local system management capabilities, remote system management using a switch module, such as the SFM6104, is also supported.

Notes

- For information regarding the location of system temperature sensors, refer to [Board Level Components](#) on page 14
 - For information on upgrading the firmware on the LDS6525, refer to the *Ensemble 6000 Intel Series with MultiCore Plus Configuration Guide*.
-

About Mercury Linux-based System Management

The LDS6525 system management functions are based on IPMI firmware provided by Pigeon Point™ Systems. The IPMI firmware:

- Provides access to the FRU EEPROM
- Monitors the module on-board sensors
- Provides access to the soft jumpers
- Accessing System Management Information From Linux

Accessing System Management Information From Linux

To collect system management information use the `ipmitool` utility. By default, `ipmitool` communicates over the KCS interface. For usage information run the `ipmitool --help` command. For a full description of this utility, visit the `ipmitool` man page on the SourceForge.net Web site at <http://ipmitool.sourceforge.net/manpage.html>.

To access the LDS6525 FRU EEPROM and the on-board sensors, you must invoke the SourceForge.net® `ipmitool` utility directly.

Mercury provides a set of `ipmitool`-based Linux scripts to disable, enable, and return the current state of the LDS6525 soft jumpers. For more information, see LDS6525 Soft Jumpers on page 67

To run the `ipmitool` utility:

From a Linux prompt, run the `ipmitool` utility using the format:

```
TOS # /usr/bin/ipmitool command
```

Table 6-1: `ipmitool` Commands for Accessing LDS6525 FRU and Sensor Information

Task	Command	Comments
Display the values of all on-board sensors	<code>sensor</code>	
Display the value of a specific on-board sensor	<code>sensor get "sensor_name"</code>	The sensor name must be inside quotation marks.
Display FRU status and configuration information	<code>fru print fru_id</code>	For a list of valid input values for <code>fru_id</code> , see Table 6-2 below

[Table 6-2](#) below lists the `fru_id` values to input to access FRU module information.

Table 6-2: Input Values for `fru_id`

Input Value	IPMI Version 0.9 and Lower	IPMI Version 0.9 and Higher
0		LDS6525 FRU
1		RTM (if installed).
2		XMC1 (if installed)
3		XMC2 (if installed)

Note

If you enter any other value for `fru_id`, the `ipmitool` command returns an error. To determine the version of IPMI you are using, refer to the `ipmitool` documentation.

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For an example of the FRU information that is available for the LDS6525, see [Example FRU Information](#) on page 101. For an example of the LDS6525 sensor thresholds, see [Board Level Components](#) on page 14.

The following examples show outputs from the `ipmitool` commands, `sensor`, `fru print`, and `sdr elist all`.

Example 6-1: Example output from the `ipmitool sensor` command.

```
[root@localhost ~]# ipmitool sensor
LDS6525 FRUState | 0x0 | discrete | 0x1080 | na | na | na | na | na | na
LDS6525 IPMB | 0x88 | discrete | 0x0880 | na | na | na | na | na | na
LDS6525 Health | 0x0 | discrete | 0x0180 | na | na | na | na | na | na
LDS6525 VSO Volt | 0x0 | discrete | 0x0180 | na | na | na | na | na | na
LDS6525 VSO Temp | 0x0 | discrete | 0x0080 | na | na | na | na | na | na
LDS6525 PLT Res | 0x0 | discrete | 0x0080 | na | na | na | na | na | na
LDS6525 PLT Sts | 0x0 | discrete | 0x0080 | na | na | na | na | na | na
LDS6525 ProcThrm | 0x0 | discrete | 0x0080 | na | na | na | na | na | na
LDS6525 tBoard1 | 42.000 | degrees C | ok | -40.000 | -37.000 | -32.000 | 85.000 | 90.000 | 95.000
LDS6525 tBoard | 36.000 | degrees C | ok | -40.000 | -37.000 | -32.000 | 75.000 | 80.000 | 85.000
LDS6525 tOutlet | 44.000 | degrees C | ok | -40.000 | -37.000 | -32.000 | 85.000 | 90.000 | 95.000
LDS6525 tFPFGA | 47.000 | degrees C | ok | -40.000 | -37.000 | -32.000 | 115.000 | 120.000 | 125.000
LDS6525 tInlet | 44.000 | degrees C | ok | -40.000 | -37.000 | -32.000 | 85.000 | 90.000 | 95.000
LDS6525 tFB-FPGA | 58.000 | degrees C | ok | -40.000 | -37.000 | -32.000 | 90.000 | 95.000 | 100.000
LDS6525 tEnet | 53.000 | degrees C | ok | -40.000 | -37.000 | -32.000 | 80.000 | 85.000 | 90.000
LDS6525 tEthPHY | 48.000 | degrees C | ok | -40.000 | -37.000 | -32.000 | 113.000 | 118.000 | 123.000
LDS6525 v12.0BP | 11.820 | Volts | ok | na | 11.460 | na | na | 12.660 | na
LDS6525 v12.0 | 11.820 | Volts | ok | na | 11.460 | na | na | 12.660 | na
LDS6525 iCPU | 4.370 | Amps | ok | na | na | na | na | 94.130 | na
LDS6525 v12.0A | 11.820 | Volts | ok | na | 11.460 | na | na | 12.660 | na
LDS6525 v-12.0A | -11.664 | Volts | ok | na | -12.554 | na | na | -11.308 | na
LDS6525 v1.8FPGA | 1.780 | Volts | ok | na | 1.710 | na | na | 1.890 | na
LDS6525 v0.67MEM | 0.670 | Volts | ok | na | 0.630 | na | na | 0.710 | na
LDS6525 vCR | 1.760 | Volts | ok | na | 1.630 | na | na | 1.810 | na
LDS6525 v1.35 | 1.330 | Volts | ok | na | 1.290 | na | na | 1.420 | na
LDS6525 v2.5 | 2.510 | Volts | ok | na | 2.380 | na | na | 2.550 | na
LDS6525 vBATTERY | 0.033 | Volts | ok | na | 2.880 | na | na | 3.161 | na
LDS6525 v5.0 | 4.920 | Volts | ok | na | 4.760 | na | na | 5.320 | na
LDS6525 v0.85 | 0.830 | Volts | ok | na | na | na | na | na | na
LDS6525 v3.3A | 3.361 | Volts | ok | na | 3.000 | na | na | 3.642 | na
LDS6525 v3.3 | 3.361 | Volts | ok | na | 3.000 | na | na | 3.642 | na
LDS6525 i12.0A | 0.000 | Amps | ok | na | na | na | na | na | na
LDS6525 v0.9 | 0.910 | Volts | ok | na | 0.850 | na | na | 0.950 | na
LDS6525 v1.8 | 1.790 | Volts | ok | na | 1.710 | na | na | 1.890 | na
LDS6525 v1.5 | 1.490 | Volts | ok | na | 1.430 | na | na | 1.580 | na
LDS6525 v1.2 | 1.140 | Volts | ok | na | 1.080 | na | na | 1.390 | na
LDS6525 v1.05 | 1.030 | Volts | ok | na | 0.950 | na | na | 1.160 | na
LDS6525 v3.3M | 3.321 | Volts | ok | na | na | na | na | na | na
LDS6525 vBATT | 1.797 | Volts | ok | na | 1.717 | na | na | 1.917 | na
LDS6525 i-12.0A | 0.000 | Amps | ok | na | na | na | na | na | na
LDS6525 v1.2FPGA | 1.190 | Volts | ok | na | 1.080 | na | na | 1.390 | na
LDS6525 v1.0FPGA | 0.990 | Volts | ok | na | 0.900 | na | na | 1.100 | na
LDS6525 v1.0 | 0.950 | Volts | ok | na | 0.900 | na | na | 1.100 | na
LDS6525 v0.67 | 0.670 | Volts | ok | na | 0.630 | na | na | 0.710 | na
LDS6525 v1.5A | 1.500 | Volts | ok | na | 1.430 | na | na | 1.580 | na
LDS6525 FS-R | na | discrete | na | na | na | na | na | na | na
LDS6525 FS-M1 | na | discrete | na | na | na | na | na | na | na
LDS6525 FS-M2 | na | discrete | na | na | na | na | na | na | na
LDS6525 tECB | 38.000 | degrees C | ok | na | na | na | 85.000 | 90.000 | 95.000
LDS6525 vECBIn | 12.100 | Volts | ok | 9.000 | 10.000 | 11.000 | 13.000 | 14.000 | 15.000
LDS6525 vECBOut | 12.100 | Volts | ok | 9.000 | 10.000 | 11.000 | 13.000 | 14.000 | 15.000
LDS6525 iECB | 4.600 | Amps | ok | na | na | na | na | 22.500 | na
LDS6525 VerChng | 0x0 | discrete | 0x0080 | na | na | na | na | na | na
```

Example 6-2: Example outputs from the `ipmitool fru print` command and output.

```
TOS# ipmitool fru print 0
Board Mfg : Mercury Systems, Inc.
Board Product :
Board Serial : XXX-nnnnn-Xn
Board Part Number : nnn-nnnnXn-nn
Board Extra : 203
Product Manufacturer : Mercury Systems, Inc.
Product Name :
Product Part Number : 910-56107-01
Product Version : 202
Product Serial : XXX-nnnnn-Xn
Product Asset Tag : MCS00000
```

Example 6-3: Example outputs from the `ipmitool fru print` command and output.

```
TOS# ipmitool fru print 1
Board Mfg : Mercury Systems, Inc.
Board Product : RTM-61P
Board Serial : JAE-00100-B2
Board Part Number : 400-1119B2-01
Board Extra : 1A
Product Manufacturer : Mercury Systems, Inc.
Product Name : RTM-61P
Product Part Number : 400-1119B2-01
Product Version : 201
Product Serial : JAE-00100-B2
Product Asset Tag : MCS000000
```

Example 6-4: Example output from the `ipmitool sdr elist all` command.

```
[root@localhost ~]# ipmitool sdr elist all
MCS LDS6525      |00h |ok |160.96 |Dynamic MC @ 82h
LDS6525 FRUState |00h |ok |160.96 |Transition to M4
LDS6525 IPMB     |01h |ok |160.96 |IPMB-A enabled, IPMB-B enabled
LDS6525 Health   |2h  |ok |160.96 |Predictive Failure Deasserted
LDS6525 VSO Volt |03h |ok |160.96 |Limit not exceeded
LDS6525 VSO Temp |04h |ok |160.96
LDS6525 PLT Res  |05h |ok |160.96
LDS6525 PLT Sts  |06h |ok |160.96
LDS6525 ProcThrm |1Ch |ok |160.96
LDS6525 tSec     |0Ch |ok |160.96 |39 degrees C
LDS6525 tPri     |0Dh |ok |160.96 |48 degrees C
LDS6525 tOut     |0Eh |ok |160.96 |49 degrees C
LDS6525 tP-FPGA  |0Fh |ok |160.96 |56 degrees C
LDS6525 tIn      |10h |ok |160.96 |39 degrees C
```

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```

LDS6525 tSwitchA      |11h |ok |160.96 |47 degrees C
LDS6525 tSwitchB      |12h |ok |160.96 |43 degrees C
LDS6525 tEth           |13h |ok |160.96 |54 degrees C
LDS6525 RTM            |00h |ok |160.96 |Local FRU @01h
LDS6525 FS-R           |07h |ok |160.96 |Transition to M4
LDS6525 FS-M1          |08h |ok |160.96 |Transition to M0
LDS6525 FS-M2          |09h |ok |160.96 |Transition to M0
LDS6525 v12.0BP        |1Dh |ok |160.96 |11.82 Volts
LDS6525 v12.0          |1Eh |ok |160.96 |11.82 Volts
LDS6525 i12.0          |1Fh |ok |160.96 |4.60 Amps
LDS6525 v12.0A         |20h |ok |160.96 |11.94 Volts
LDS6525 v-12.0A        |21h |ok |160.96 |-17.00 Volts
LDS6525 v1.8A          |22h |ok |160.96 |1.77 Volts
LDS6525 vCR            |23h |ok |160.96 |0.90 Volts
LDS6525 vAXG           |24h |ok |160.96 |0 Volts
LDS6525 vCCSA          |25h |ok |160.96 |0.89 Volts
LDS6525 v2.5           |26h |ok |160.96 |2.48 Volts
LDS6525 v5.0           |27h |ok |160.96 |4.92 Volts
LDS6525 i3.3A          |28h |ok |160.96 |0.20 Amps
LDS6525 v3.3A          |29h |ok |160.96 |3.32 Volts
LDS6525 v3.3           |2Ah |ok |160.96 |3.32 Volts
LDS6525 i12.0A         |2Bh |ok |160.96 |0 Amps
LDS6525 v0.9           |2Ch |ok |160.96 |0.91 Volts
LDS6525 v1.8           |2Dh |ok |160.96 |1.80 Volts
LDS6525 v1.5           |2Eh |ok |160.96 |1.51 Volts
LDS6525 v1.2           |2Fh |ok |160.96 |1.19 Volts
LDS6525 v1.05          |30h |ok |160.96 |1.04 Volts
LDS6525 v3.3M          |31h |ok |160.96 |3.32 Volts
LDS6525 vBATT          |32h |ok |160.96 |1.80 Volts
LDS6525 i-12.0A        |33h |ok |160.96 |0 Amps
LDS6525 v1.2A          |34h |ok |160.96 |1.18 Volts
LDS6525 v1.0A          |35h |ok |160.96 |0.99 Volts
LDS6525 v1.0           |36h |ok |160.96 |0.98 Volts
LDS6525 v0.75          |37h |ok |160.96 |0.74 Volts
LDS6525 v1.5A          |38h |ok |160.96 |1.49 Volts

```

Temperature Monitoring

The temperature sensors on the LDS6525 are programmed to generate events when thresholds are exceeded. Unless the Environmental Bypass signal from the backplane is asserted, when the upper non-recoverable threshold is exceeded on any temperature sensor, the module's payload power is turned off, however the system management circuitry remains powered on.

Table 6-3, below, lists the temperature sensors on the LDS6525.

Table 6-3: Temperature Sensors

Sensor	Location
LDS6525 tBoard1	Near onboard battery
LDS6525 tBoard	Near Inlet
LDS6525 tOutlet	At air outlet
LDS6525 tP-FPGA	Internal to tP-FPGA
LDS6525 tInlet	Near onboard battery
LDS6525 tFB-FPGA	Internal to tFB-FPGA
LDS6525 tEnet	Center of Board
LDS6525 tEthPHY	Internal to tEthPHY
LDS6525 tECB	Near ECB circuit

Table 2-8 below lists the temperature sensor thresholds on the LDS6525.

Table 6-4: Air-cooled Temperature Sensor Thresholds

Sensor	Lower Non-Recoverable Threshold	Lower Critical Threshold	Lower Non-Critical Threshold	Upper Non-Critical Threshold	Upper Critical Threshold	Upper Non-Recoverable Threshold
LDS6525 tBoard1	-40.000 C	-37.000 C	-32.000 C	85.000 C	90.000 C	95.000 C
LDS6525 tBoard	-40.000 C	-37.000 C	-32.000 C	75.000 C	80.000 C	85.000 C
LDS6525 tOutlet	-40.000 C	-37.000 C	-32.000 C	85.000 C	90.000 C	95.000 C
LDS6525 tP-FPGA	-40.000 C	-37.000 C	-32.000 C	115.000 C	120.000 C	125.000 C
LDS6525 tInlet	-40.000 C	-37.000 C	-32.000 C	85.000 C	90.000 C	95.000 C
LDS6525 tFB-FPGA	-40.000 C	-37.000 C	-32.000 C	90.000 C	95.000 C	100.000 C
LDS6525 tEnet	-40.000 C	-37.000 C	-32.000 C	80.000 C	85.000 C	90.000 C
LDS6525 tEthPHY	-40.000 C	-37.000 C	-32.000 C	113.000 C	118.000 C	123.000 C
LDS6525 tECB	N/A	N/A	N/A	85.000 C	90.000 C	95.000 C

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Table 2-9 below, lists the signals from the processor that, when asserted for the ProcThrm discrete sensor, indicate an over-temperature condition.

Table 6-5: Processor Thermal Management Signals

Signal	Assertion Condition
PROCHOT1_N	Any CPU core reaches maximum operating temperature
TEMP_ALERT_N	CPU or PCH reaches temperature outside upper or lower limit set by BIOS
PCHHOT1_N	PCH reaches temperature greater than limit set by BIOS
IPMC_PROCHOT1_N	M-FPGA IPMC output to indicate the CPU to cool down

The processor temperature values can be read from the M-FPGA IPMC through the PCH SMLink1.

Voltage Monitoring

The voltage sensors on the LDS6525 are programmed to generate events when thresholds are exceeded. When the upper non-recoverable threshold is exceeded on any voltage sensor, the module is *not* turned off. Sensor thresholds are listed in [Sensor Data Records](#) on page 1.

Table 6-6 below lists the voltage sensors on the LDS6525.

Table 6-6: LDS6525Voltage Sensors

Sensor	Nominal Voltage, VAC	Description
LDS6525 v12.0BP	12 volts	Voltage sensor for the 12 volt Backplane rail
LDS6525 v12.0	12 volts	Voltage sensor for the 12 volt rail
LDS6525 v12.0A	12 volts	Voltage sensor for the Auxiliary 12 volt rail
LDS6525 v-12.0A	-12 volts	Voltage sensor for the -12 volt rail
LDS6525 v1.8FPGA	1.8 volts	Voltage sensor for the 1.8 volt rail
LDS6525 v0.67MEM	0.67 volts	Voltage sensor for the 0.67 volt rail
LDS6525 vCR	1.8 volts	Voltage sensor for the 1.8 volt Core Reuse Block rail
LDS6525 v1.35	1.35 volts	Voltage sensor for the 1.35 volt rail
LDS6525 v2.5	2.5 volts	Voltage sensor for the 2.5 volt rail
LDS6525 vBATTERY	3.0 volts	Voltage sensor for the 3.0 volt Battery rail
LDS6525 v5.0	5.0 volts	Voltage sensor for the 5.0 volt rail
LDS6525 v0.85	0.85 volts	Voltage sensor for the 0.85 volt rail

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Sensor	Nominal Voltage, VAC	Description
LDS6525 v3.3A	3.3 volts	Voltage sensor for the Auxiliary 3.3 volt rail
LDS6525 v3.3	3.3 volts	Voltage sensor for the 3.3 volt rail
LDS6525 v0.9	0.9 volts	Voltage sensor for the 0.9 volt rail
LDS6525 v1.8	1.8 volts	Voltage sensor for the 1.8 volt rail
LDS6525 v1.5	1.5 volts	Voltage sensor for the 1.5 volt rail
LDS6525 v1.2	1.2 volts	Voltage sensor for the 1.2 volt rail
LDS6525 v1.05	1.05 volts	Voltage sensor for the 1.05 volt rail
LDS6525 v3.3M ail	3.3 volts	Voltage sensor for the Mezzanine 3.3 volt r
LDS6525 vBATT	1.8 volts	Voltage sensor for the 1.8 volt Battery rail
LDS6525 v1.2FPGA	1.2 volts	Voltage sensor for the FPGA 1.2 volt rail
LDS6525 v1.0FPGA s	1.0 volt	Voltage sensor for the FPGA 1.0 volt rail
LDS6525 v1.0	1.0 volts	Voltage sensor for the 1.0 volt rail
LDS6525 v0.67	0.67 volts	Voltage sensor for the 0.67 volt rail
LDS6525 v1.5A	1.5 volts	Voltage sensor for the Auxiliary 1.5 volt rail
LDS6525 vECBIn	12 volts	Voltage sensor for the backplane 12 volt rail before the ECB
LDS6525 vECBOut	12 volts	Voltage sensor for the backplane 12 volt rail after the ECB

Table 6-7: LDS6525 Voltage Sensor Thresholds

Sensor	Lower Non-Recoverable Threshold	Lower Critical Threshold	Lower Non-Critical Threshold	Upper Non-Critical Threshold	Upper Critical Threshold	Upper Non-Recoverable Threshold
LDS6525 v-12.0A	NA	-12.554	NA	NA	-11.308	NA
LDS6525 v-0.67	NA	0.630	NA	NA	0.710	NA
LDS6525 v0.67MEM	NA	0.630	NA	NA	0.710	NA
LDS6525 v0.85	NA	NA		NA	NA	NA
LDS6525 v0.9	NA	0.850	NA	NA	0.950	NA
LDS6525 v1.0	NA	0.900	NA	NA	1.100	NA
LDS6525 v1.05	NA	0.950	NA	NA	1.160	NA

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Sensor	Lower Non-Recoverable Threshold	Lower Critical Threshold	Lower Non-Critical Threshold	Upper Non-Critical Threshold	Upper Critical Threshold	Upper Non-Recoverable Threshold
LDS6525 v1.0FPGA	NA	0.900	NA	NA	1.100	NA
LDS6525 v1.2	NA	1.080	NA	NA	1.390	NA
LDS6525 v1.2FPGA	NA	1.080	NA	NA	1.390	NA
LDS6525 v1.35	NA	1.290	NA	NA	1.420	NA
LDS6525 v1.5	NA	1.430	NA	NA	1.580	NA
LDS6525 v1.5A	NA	1.430	NA	NA	1.580	NA
LDS6525 v1.8	NA	1.710	NA	NA	1.890	NA
LDS6525 v1.8FPGA	NA	1.710	NA	NA	1.890	NA
LDS6525 v12.0	NA	11.460	NA	NA	12.660	NA
LDS6525 v12.0A	NA	11.460	NA	NA	12.660	NA
LDS6525 v12.0BP	NA	11.460	NA	NA	12.660	NA
LDS6525 v2.5	NA	2.380	NA	NA	2.550	NA
LDS6525 v3.3	NA	3.000	NA	NA	3.642	NA
LDS6525 v3.3A	NA	3.000	NA	NA	3.642	NA
LDS6525 v3.3M	NA	NA	NA	NA	NA	NA
LDS6525 v5.0	NA	4.760	NA	NA	5.320	NA
LDS6525 vBATT	NA	1.717	NA	NA	1.917	NA
LDS6525 vBATTERY	NA	2.880	NA	NA	3.161	NA
LDS6525 vCR	NA	1.630	NA	NA	1.810	NA
LDS6525 vECBIn	9.000	10.000	11.00	13.000	14.000	15.000
LDS6525 vECBOut	9.000	10.000	11.00	13.000	14.000	15.000

Current Monitoring

The current sensors on the LDS6525 are programmed to generate events when thresholds are exceeded. When the upper non-recoverable threshold is exceeded on any current sensor, the module is *not* turned off. Sensor thresholds are listed in [Sensor Data Records](#) on page 1.

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Table 6-8 below lists the current sensors on the LDS6525.

Table 6-8: LDS6525Current Thresholds

Sensor	Lower Non-Recoverable Threshold	Lower Critical Threshold	Lower Non-Critical Threshold	Upper Non-Critical Threshold	Upper Critical Threshold	Upper Non-Recoverable Threshold
LDS6525 i-12.0A	NA	NA	NA	NA	NA	NA
LDS6525 i12.0A	NA	NA	NA	NA	NA	NA
LDS6525 iCPU	NA	NA	NA	NA	94.130	NA
LDS6525 iECB	NA	NA	NA	NA	22.500	NA

LDS6525 Soft Jumpers

Table 6-9 below lists the LDS6525 soft jumper settings and functions. Any soft jumpers not listed are for factory use only.

Table 6-9: Soft Jumper Settings

Device ID	Bit Position	Jumper	Function	Setting	Result
0	0	FP_UART_SELECTED	Configures whether the IPMC or the CPU has access to the micro DB-9 port B interface.	Clear	The processor has access to the front panel micro DB-9 port B interface, and the IPMC has access to the port D UART interface on the backplane.
				Set (Default)	The processor has access to the port D UART interface on the backplane, and the IPMC has access to the front panel micro DB-9 port B interface.
0	1	BP_UARTA_RS422_SELECTED	Switches the UART-A signals between RS-232 and RS-422/RS-485 levels. When the soft jumper is set to RS-422/RS-485 levels, UART flow control is no longer available.	Clear (Default)	Enables RS-232 mode.
				Set (Default)	Enables RS-422/RS-485 mode.

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Device ID	Bit Position	Jumper	Function	Setting	Result
0	2	BP_UARTB_RS422_SELECTED	Switches the UART-B signals between RS-232 and RS-422/RS-485 levels. When the soft jumper is set to RS-422/RS-485 levels, UART flow control is no longer available.	Clear (Default)	Enables RS-232 mode.
				Set (Default)	Enables RS-422/RS-485 mode.
0	3	FP_ETH_SELECTED	Configures whether one of the two 10/100/1000BASE-T ports is accessible via the front panel or the backplane.	Clear (Default)	10/100/1000BASE-T port goes to the backplane.
				Set (Default)	10/100/1000BASE-T port goes to the front panel.
0	4	UNDEFINED	UNDEFINED	N/A	N/A
				N/A	N/A
0	5	MM_RESET_IGNORED	Configures whether the module responds to the assertion and de-assertion of the backplane MM reset signal.	Clear	Masks the backplane MM reset signal. The module will not respond to the assertion or de-assertion of the backplane MM reset signal.
				Set (Default)	Enables the backplane MM reset signal. The module will respond to the assertion or de-assertion of the backplane MM reset signal.
1	0	FB-FPGA_CFG_SEL0	The FB_FPGA_CFG_SEL[2:0] soft jumpers select which of the four bitstreams will be loaded during configuration of the FB-FPGA. These soft jumpers are inputs to the P-FPGA. Setting these jumpers does not initiate loading of a bitstream.	For information regarding FB_FPGA_CFG_SEL Soft Jumper Definitions, see Table 6-10 on page 71 For information regarding FPGA_CFG_SEL functions, see Table 6-11 on page 72	
1	1	FB-FPGA_CFG_SEL1			
1	2	FB-FPGA_CFG_SEL2			

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Device ID	Bit Position	Jumper	Function	Setting	Result
1	3	UNDEFINED	N/A	N/A	N/A
				N/A	N/A
1	4	ME_DISABLED	The ME_DISABLE soft jumper is used to set control the HAD_SDO strap on the PCH.	Set	ME Disabled (MS Legacy Mode, BIOS must match).
				Clear (Default)	ME Enabled (BIOS must match) [Default].
1	5	P-FPGA_BOOT_RAM_ENABLED	The P-FPGA_BOOT_RAM_ENABLED soft jumper is used to enable the boot process and operation to use SRAM in conjunction with the SPI Flash. Table 110 shows the definition of the SJ_PFPGA_BOOT_RAM soft jumper.	Set	Enable boot process and operation with SRAM support
				Clear	Normal boot processes and operation using SPI Flash. [Default]
2	1	P-FPGA_EEPROM_DISABLED	The P-FPGA_EEPROM_DISABLED soft jumper is used to inhibit configuration of the P-FPGA from EEPROM. The jumper is typically employed if the configuration data in the EEPROM has been corrupted. Corrupted EEPROM data could cause unexpected behavior.	Set	Inhibits configuration of the PLX from EEPROM
				Clear	Allows configuration of the PLX from EEPROM. [Default]

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Device ID	Bit Position	Jumper	Function	Setting	Result
2	2	P-FPGA_CFG_SEL0	The P-FPGA_CFG_SEL1 and P-FPGA_CFG_SEL0 soft jumpers select which one of four bitstreams will be loaded during 4090 configuration of the P-FPGA. These soft jumpers are connected to the P-FPGA FLASH address pins A23 and A24 via mux with 4091 the P-FPGA's RS[0:1] bits. Setting these jumpers does not initiate loading of a bitstream.	For more information, see Table 6-12 on the next page	
2	3	P-FPGA_CFG_SEL1			
2	4	PMC_100M_SELECTED	The SEL_PMC_100M soft jumper is used to configure the PCI-X bus frequency for the module's PMC site to 100MHz. This 4147 jumper should only be set when the PMC site is requesting a bus frequency of 133MHz. Setting this jumper at other frequencies 4148 will result in a downgrading of the requesting frequency (from 66MHz to 50MHz or from 33MHz to 25MHz).	Set	
				Clear	
2	5	UNDEFINED	N/A	N/A	N/A
				N/A	N/A

Table 6-10: FB-FPGA_CFG_SEL Soft Jumper Definitions

Jumper	Device #	Jumper #
FB-FPGA_CFG_SEL0	1	0
FB-FPGA_CFG_SEL1	1	1
FB-FPGA_CFG_SEL2	1	2

Table 6-11: FB-FPGA_CFG_SEL State Table

*_CFG_SEL2	*_CFG_SEL1	*_CFG_SEL0	Function
Set	N/A	N/A	Do not load FB-FPGA
Clear	Set	Set	Configures FB-FPGA from bitstream 3 (failsafe - write protected by hardjumper)
Clear	Set	Clear	Configures FB-FPGA from bitstream 2
Clear	Clear	Set	Configures FB-FPGA from bitstream 1
Clear	Clear	Clear	Configures FB-FPGA from bitstream 0 (fallback bitstream)

Table 6-12: P-FPGA_CFG_SEL Soft Jumper Definitions

Jumper	Device #	Jumper #
P-FPGA_CFG_SEL0	2	2
P-FPGA_CFG_SEL1	2	3

Table 6-13: P-FPGA_CFG_SEL State Table

P-FPGA_CFG_SEL1	P-FPGA_CFG_SEL0	Function
Set	Set	Configures P-FPGA from bitstream 3 (failsafe - write protected by hardjumper)
Set	Clear	Configures P-FPGA from bitstream 2
Clear	Set	Configures P-FPGA from bitstream 1
Clear	Clear	Configures P-FPGA from bitstream 0 (fallback bitstream)

FPGA Bitstream Selection

The `FPGA_CFG_SEL1` and `FPGA_CFG_SEL0` soft jumpers select one of four available bitstreams to load during P-FPGA configuration. Setting these jumpers does not initiate bitstream loading.

Table 6-14: LDS6525 FPGA Bitstream Selection

FPGA_CFG_SEL1	FPGA_CFG_SEL0	Function
Set	Set	Configures P-FPGA from bitstream 3
Set	Clear	Configures P-FPGA from bitstream 2

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FPGA_CFG_SEL1	FPGA_CFG_SEL0	Function
Clear	Set	Configures P-FPGA from bitstream 1
Clear	Clear	Configures P-FPGA from bitstream 0

Soft Jumper Scripts

Table 6-15 below lists the scripts you can use to access and manipulate the LDS6525 soft jumpers.

Table 6-15: Soft Jumper Scripts

Script	Description
<code>mc_module_set_soft_jumper.sh</code>	Sets the soft jumper device value.
<code>mc_module_get_soft_jumper.sh</code>	Gets the soft jumper device value.
<code>mc_module_set_bitstream_partition.sh</code>	Sets the bitstream partition.

Setting a Soft Jumper

This example provides the jumper device ID and the jumper bit positions for use with the soft jumper scripts.

The script sets:

- soft jumper device 0
- bit 3
- value of 1 (set)

<code>\$ /mercury/bin/mc_module_set_soft_jumper.sh 0 3 1</code>
<code>mc_module_set_soft_jumper.sh: device 0 bit 3/ETH_FP_SEL operation SET success</code>

Reading a Soft Jumper

These examples illustrate the script reading the soft jumper setting.

The script reads:

- soft jumper device 0
- bit 3

<code>\$ /mercury/bin/mc_module_get_soft_jumper.sh 0 3</code>
<code>mc_module_get_soft_jumper.sh: device 0 bit 3/ETH_FP_SEL is SET</code>
<code>mc_module_get_soft_jumper.sh: device 0 bit 3/ETH_FP_SEL is CLEAR</code>

System Signals

The LDS6525 uses the following system signals.

System Reset

When asserted, this system signal brings the OpenVPX plug-in modules to a well-defined normal condition or initial state.

Non-volatile Memory Read Only (NVMRO)

When asserted, this system signal prevents system users from modifying the contents of all software-accessible Non-volatile memory on an OpenVPX plug-in module. When NVMRO is asserted, the system users still have read access to the non-volatile memory contents.

The NVMRO signal is read once, when the module first powers up. You must reset the module to have it see a change in the NVMRO signal.

The existence of an NVMRO signal on a plug-in module enables systems to run sensitive applications (when NVMRO is enabled) or handle sensitive data without the concern that the data is kept after power is removed. The signal also ensures that no sensitive information can be stored (intentionally or inadvertently) in persistent memory for later retrieval.

Environmental Bypass

When asserted, this system signal indicates that the system will attempt to continue to operate normally instead of responding to any future environmental events (that occur while the Environmental Bypass signal is still asserted) that would normally result in shutdown of an individual module, groups of modules, or the entire system. As an example, this signal, when asserted, disables automatic shutoff of the LDS6525 when the critical upper temperature limits on the CRB thermal sensors are exceeded.

This behavior is helpful in OpenVPX system implementations where continued system operation is deemed more important than preventing possible module damage. For example, if an OpenVPX plug-in module is implemented to power-down with an over-temperature event, an asserted Environmental Bypass signal can be used on this module to mask the over temperature event. This, in turn, enables the module to continue to operate instead of powering down.

Chassis Test

When asserted, this OpenVPX system signal indicates that the system will discontinue normal operation and instead begin executing extensive system-wide diagnostics. This signal acts the same as SYS_RESET, except the POST runs in extended mode when it is asserted.

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This behavior is helpful in OpenVPX systems to aid in the diagnosis and resolution of system failures that can be isolated to failed field replaceable units (FRUs). For example, an operator that detects a fault in the system that should be isolated to the FRU level could initiate a Chassis Test on the system.

Module Maskable Reset

When asserted, this system signal brings OpenVPX plug-in modules to a well-defined normal condition or initial state. This functional behavior is the same as the behavior resulting from the assertion of the System Reset signal. Module Maskable Reset functionally differs from the System Reset in that you can configure it per individual receiver to either monitor or ignore the signal assertion.

This functionality allows OpenVPX system integrators the ability to synchronously reset a subdivision of their chassis contents while the remainder of the chassis continues to operate. For example, a system might require the ability to reset the front-end digital signal processing system while keeping the control functions running.

Memory Clear

When asserted, this system signal indicates the desire of the system to clear all system memory of any potentially sensitive information. This includes both volatile and Non-volatile memory that can be accessed by system software. The signal is monitored but the system takes no independent action. This behavior is helpful in OpenVPX system implementations to prevent proprietary information from being inappropriately accessed.

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Chapter 7

Interacting with the BIOS

Note

The screen shots shown herein are representative only and might not reflect the shipping version of BIOS Setup screens for your device.

Terminal-specific Settings

Depending on the terminal tool used, to manage BIOS functions, terminal settings often have to be modified so that the BIOS setup, option ROM, and OS loader menus properly display information and respond to keystrokes.

For example:

- When using *TeraTerm* You must select the *Full Color* option. Otherwise, most items that are highlighted in the BIOS menu are not visible.
- Flow control should be set to *None*, or *Software*.
- The serial port settings should be set to *115200 bps, 8 bits data, no parity, 1 stop bit* (i.e., "115200,8,N,1).
- When using *PuTTY*, You must change the backspace key from CTRL-? To CTRL-H.

BIOS Information Screen

When you power-up or reset the LDS6525, the BIOS Splash screen displays the following information ([Figure 7-1](#) on the next page).

This screen displays each firmware component version with:

- The brand string
- The board revision screen
- The SPI Flash configuration
- The current BIOS firmware version

- The current BIST version
- The current FPGA firmware versions

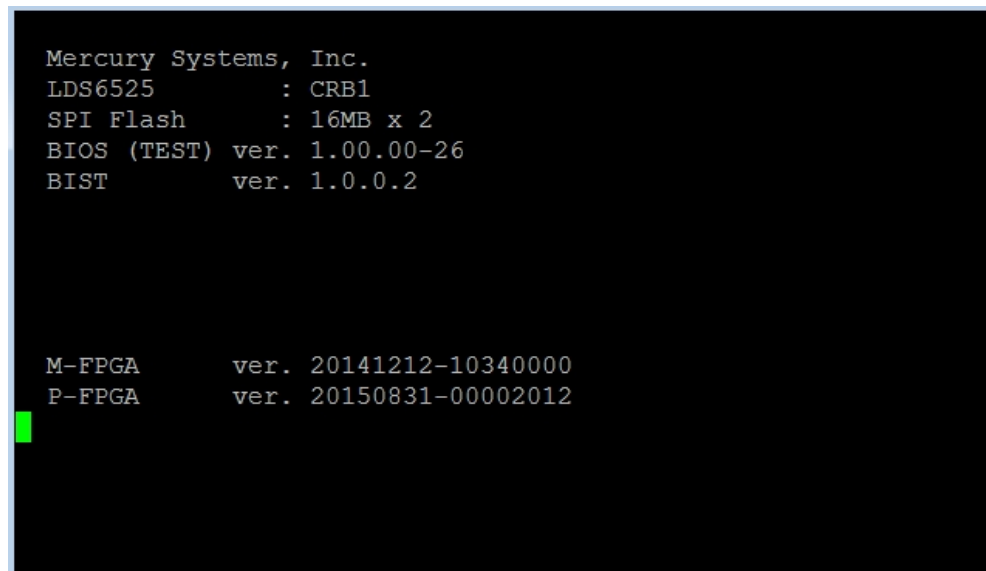


Figure 7-1: Initial BIOS Splash screen

After the BIOS completes system initialization, a second information screen ([Figure 7-2](#) below) displays:

- Copyright string
- BIOS reference release date
- BIOS reference version

This second screen is briefly displayed (1 to 2 seconds) during which you may press the <DELETE> or <ESC> key to enter the BIOS menu.

If you do not press a key while this screen is displayed, the BIOS continues to boot using the default boot device at the top of the boot order.

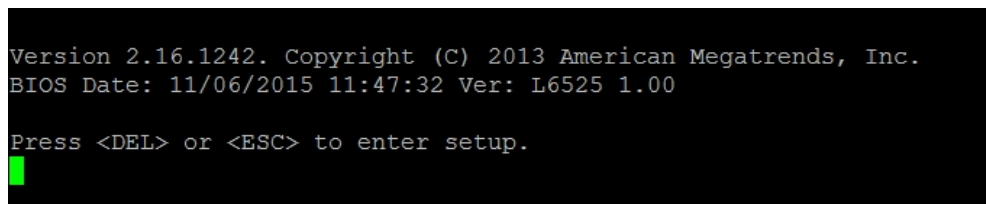


Figure 7-2: Second BIOS Splash screen

BIOS Main Menu

After pressing the <Esc> or keys at the second Splash screen (Figure 7-2 on the previous page), the scrollable main menu (Figure 7-3 below) displays.

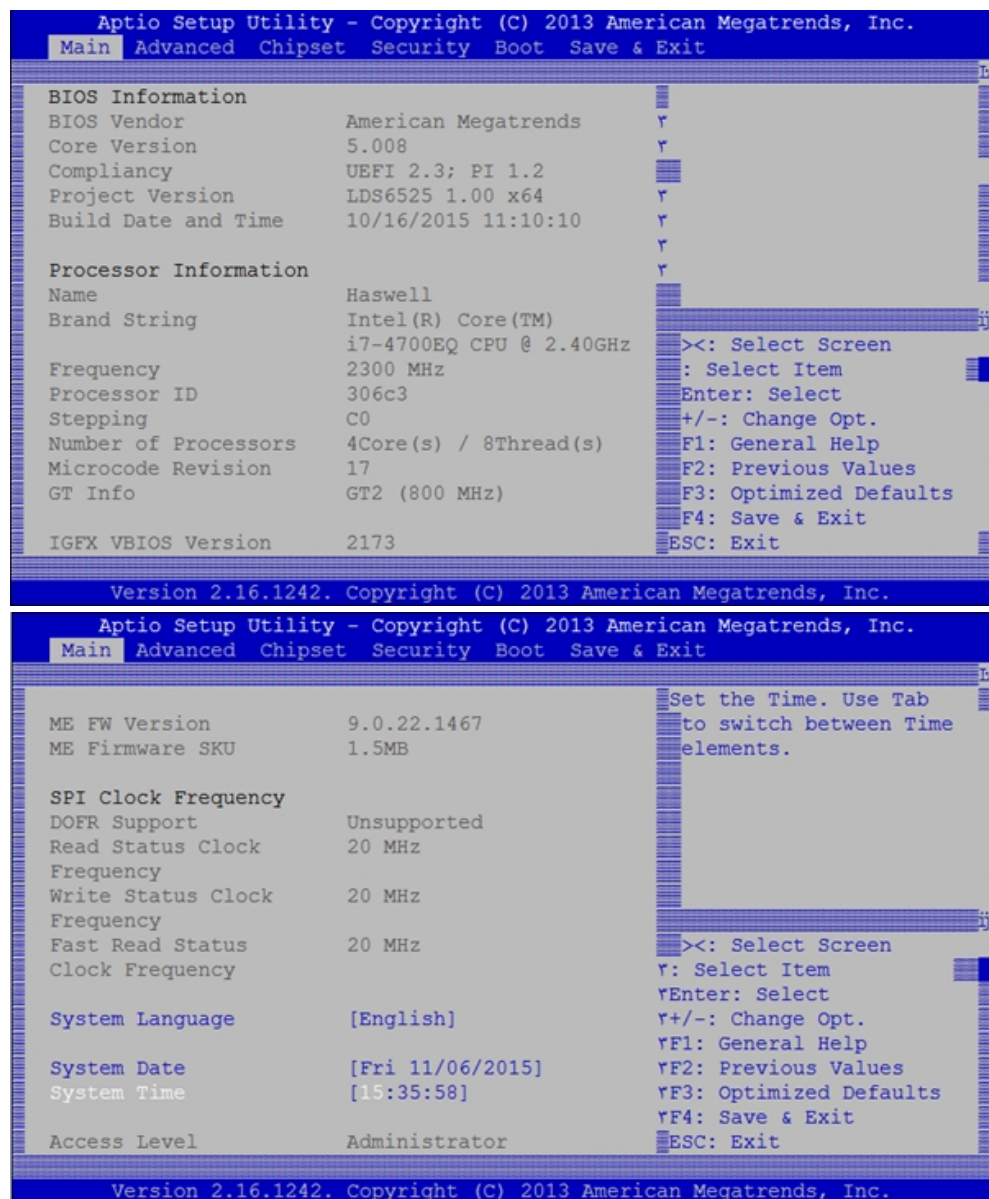


Figure 7-3: BIOS Main Menu

Exiting the BIOS Menu

To exit the BIOS Menu, from the menu bar, select *Save and Exit* (Figure 7-4 below).

- To continue booting, but ignoring any changes, select *Discard Changes and Exit*.
- To continue booting, and saving all changes, select *Save Changes and exit*.
- To reset the module automatically so that all changes take effect, select *Save Changes and Reset*.

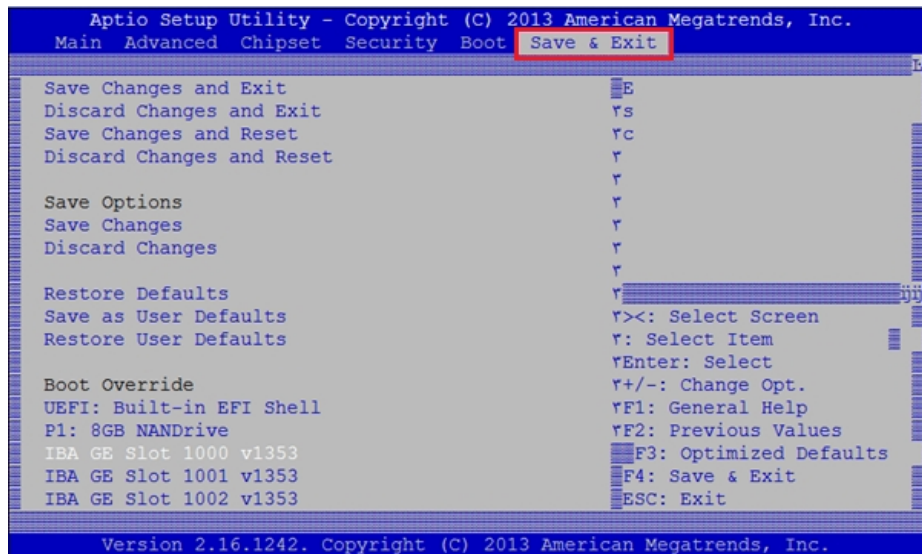


Figure 7-4: Using the BIOS *Save & Exit* Menu

Notes:

- If you select *Save Changes and Exit*, changes made will not take effect until the module is reset.
- If you save changes with *Save Changes*, and then exit using *Exit Discarding Changes*, changes will not take effect until the module is reset.
- To use *Save Changes and Exit* without navigating to the exit screen, press the <F4> key from any BIOS Menu screen.
- To return all values to the defaults at any point, press the <F3> key.
- To return to the previous screen at any point, press the <Esc> key.
- Some settings are required to force a reset after you exit Setup, even if you choose *Save Changes and Exit* instead of *Save Changes and Reset*.

Setting BIOS Passwords

The BIOS provides supervisor and user-level password protection from unauthorized system boot and Setup Utility changes.

After you set the passwords, when the module resets, the password prompt is displayed after a sign-on message. The supervisor or user password must be correctly entered before booting can proceed.

Keep in mind the following guidelines when setting the BIOS password:

- If only the administrator password is set, it will limit access to the BIOS menu and is asked only when entering the BIOS menu.
- If only the User password is set, then it is used as a power-on password and must be entered to boot or enter the BIOS menu. Once in the BIOS menu, the user will have Administrator rights.
- If both passwords are set, then either password can be used as a power on password. If the administrator password is used, the user will have Administrator rights in the BIOS menu. If the user password is used, the user will have User rights in the BIOS menu.
- Administrator rights in the BIOS menu allows for full access to all options, where User rights only allow for changing the boot order.

To set the BIOS password:

1. Select *Administrator Password* or *User Password*(Figure 7-5 below).



Figure 7-5: Setting a password

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2. If a password is required, at the password prompt ([Chapter 7](#) on page 77, below) enter the password and hit <Enter>.
3. At the password confirmation prompt, enter the password and hit <Enter>.

Note: To uninstall the supervisor or user password, set the password entering no characters (leave password field blank). The module BIOS resets to the new password value once you save your settings and exit the BIOS Menu.

Configuring the BIOS

The following BIOS configuration tasks are supported on the LDS6525 module:

- Overriding the Default Boot Device (for the current Boot session only)
- Changing the Boot Order
- Controlling Hyper-Threading
- Controlling BIST Behavior

Overriding the Default Boot Device

The BIOS Menu displays a list of bootable devices and provides a means to override the default boot device. The following procedure demonstrates how to boot from a legacy PXE server through the front panel network interface, instead of allowing the internal SATA drive to boot by default.

To override the Default Boot Device:

1. In the BIOS menu, select the *Save & Exit* tab (Figure 7-6 below, below).
2. Select the desired boot device and then press *Enter* to begin booting the selected device.

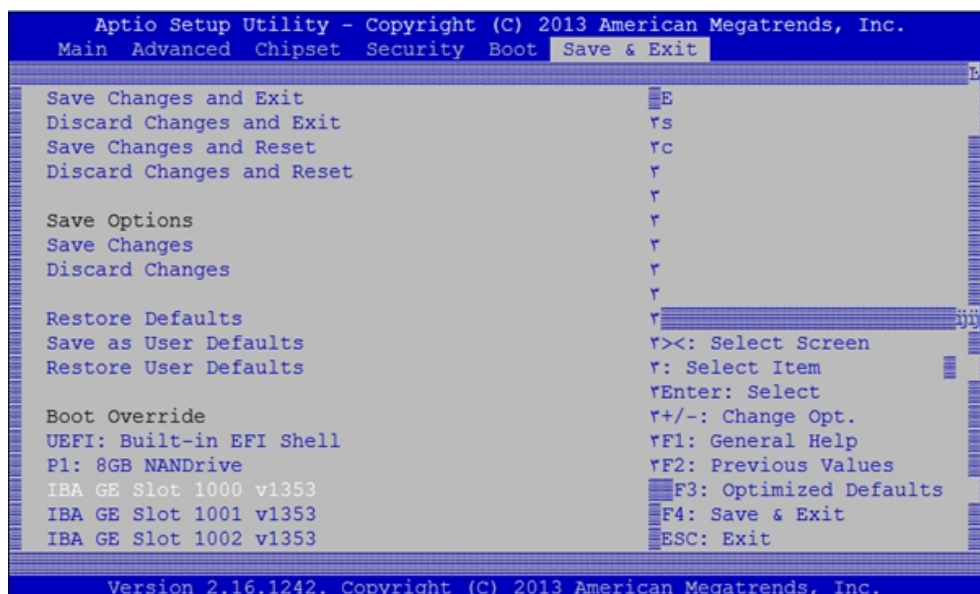


Figure 7-6: Selecting a Boot Device from the Save & Exit Tab

Changing the Boot Order for All Future Boot Sessions

The BIOS attempts to boot the first device in the boot order list, and, if unable to do so, attempts to boot the second device, and so on until the boot list is exhausted.

The boot device order is managed first according to *Boot Option Priority*, and then by individual devices within each boot option. If there is more than one device within a boot option, you may change its priority within its *Boot Option Priority*.

Note: The boot priority changes will take effect the next time the module is reset.

To change any Boot Option Priority:

1. In the BIOS menu, select the *Boot* tab.
2. Use the up/down keys to select *.(Figure 7-7 below).*
3. Press <Enter> to display boot device options ().

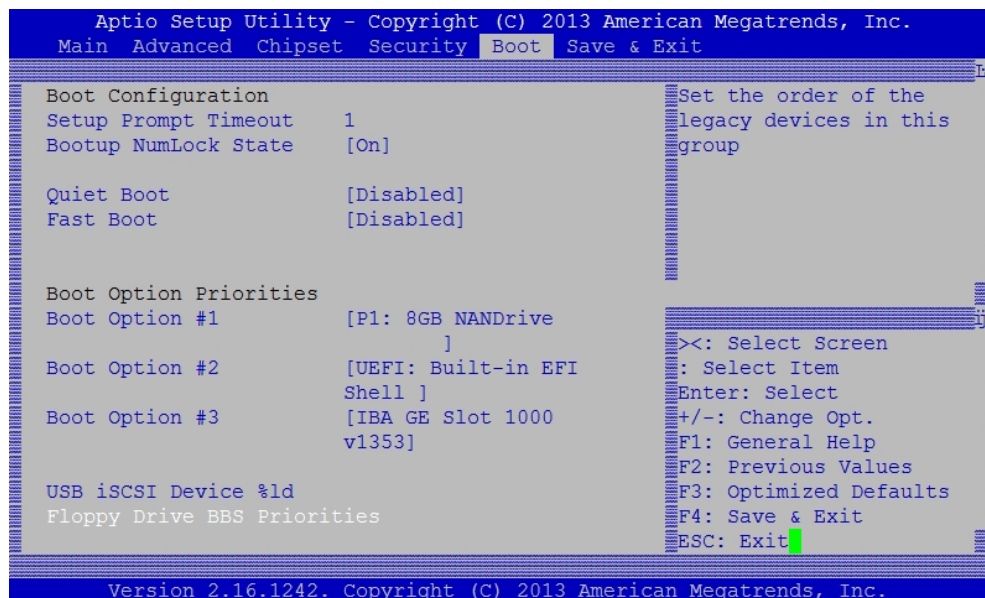


Figure 7-7: Defining the Boot Device Priority

To select a new device within a device type:

1. In the Boot menu, use the up/down arrows to select the Boot Option, and press <Enter> to display the available devices (Figure 7-8 on the facing page).
2. Use the up/down arrows to select the desired boot device and press <Enter>.

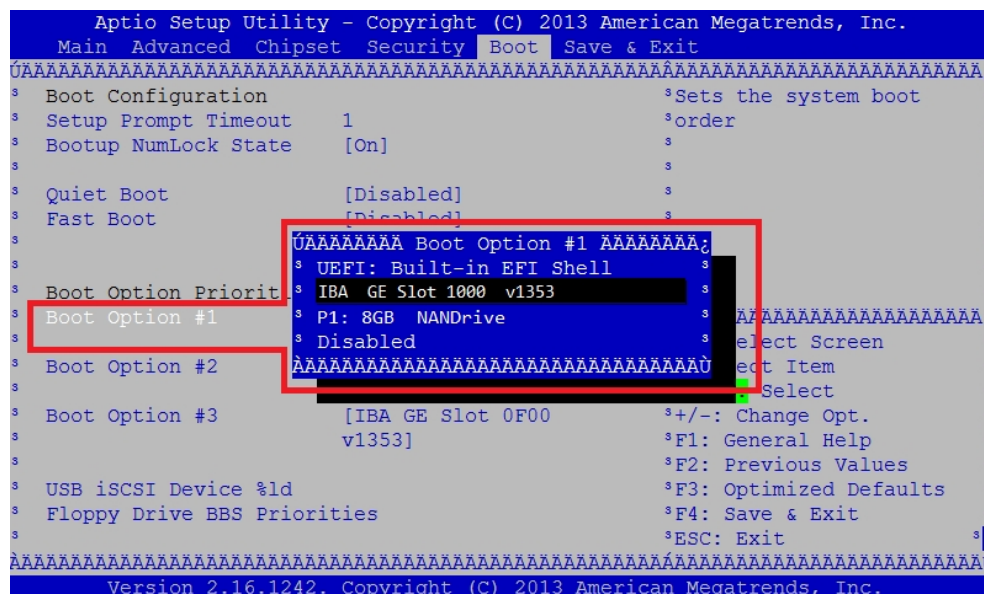


Figure 7-8: Selecting a new Device

The newly-selected device now displays in the Boot Option 1 position (Chapter 7 on page 77).

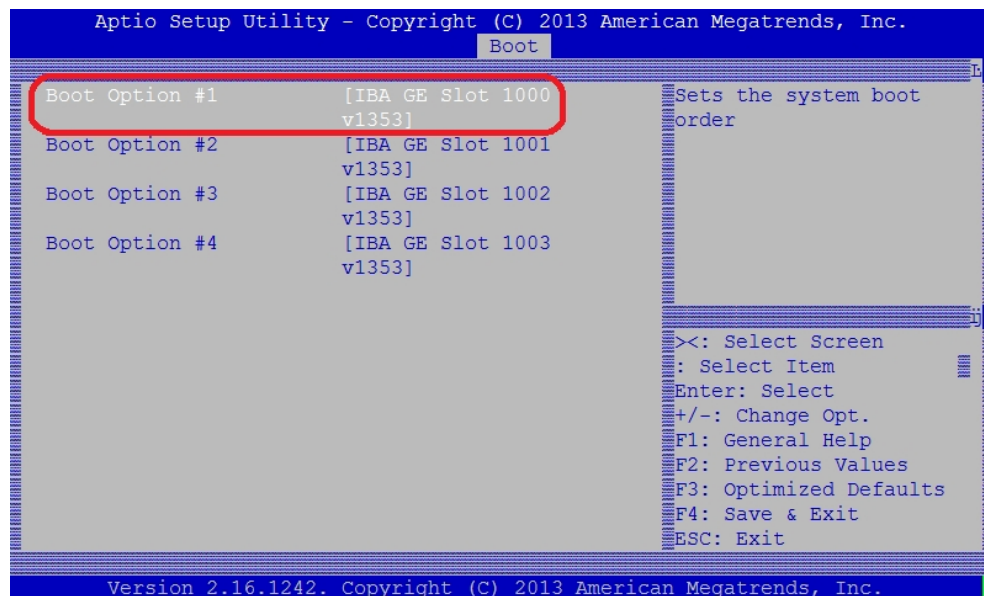


Figure 7-9: Selected Boot Option

Controlling Hyper-Threading

The LDS6525BIOS Hyper-threading feature allows each cpu core to run two simultaneous pipelined sets of instructions in parallel. Each HT thread has an independent set of cpu general purpose registers, integer execution unit, interrupt controller, and cpu state registers; shared resources include the prefetcher, branch prediction unit, and first-level caches.

For certain classes of operations, an HT thread can gain as much additional throughput as an additional traditional core. Operations in this class include highly I/O-bound operations, or very small but fast loops which live in the first-level caches and therefore do not exercise the branch prediction or prefetcher logic.

Generally, Mercury recommends enabling HT (by default) on LDS-class boards, where I/O-based data collection is typically the priority.

To change the state of Hyper-Threading:

1. In the BIOS Menu, select the *Advanced* tab ([Chapter 7](#) on page 77).
2. On the Advanced tab, select *CPU Configuration* ([Figure 7-10](#) below).

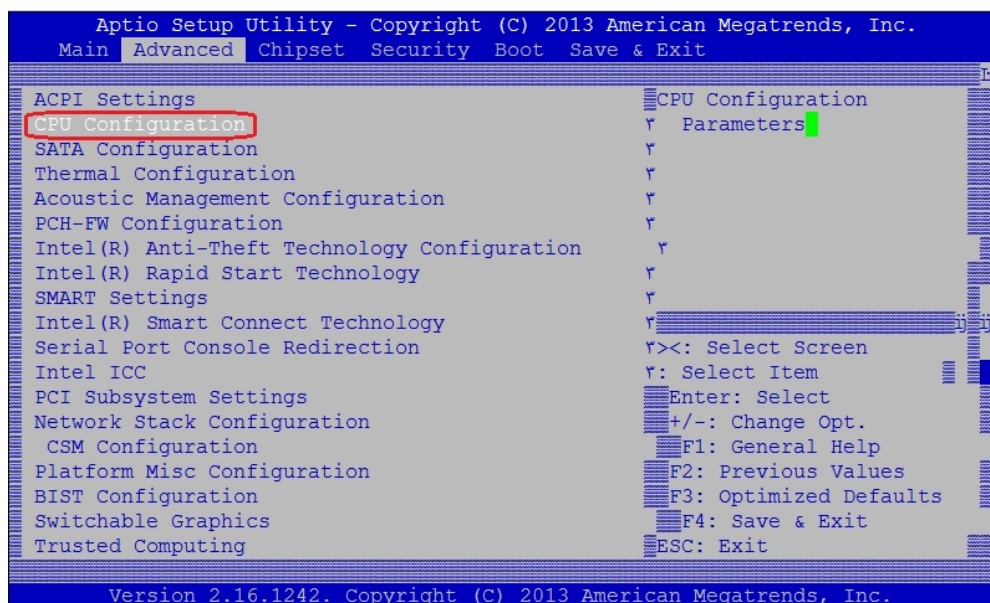


Figure 7-10: Selecting CPU Configuration

3. On the CPU Configuration screen, select *Hyper-threading*, and then select *Enabled* or *Disabled* ([Figure 7-11](#) on the facing page).

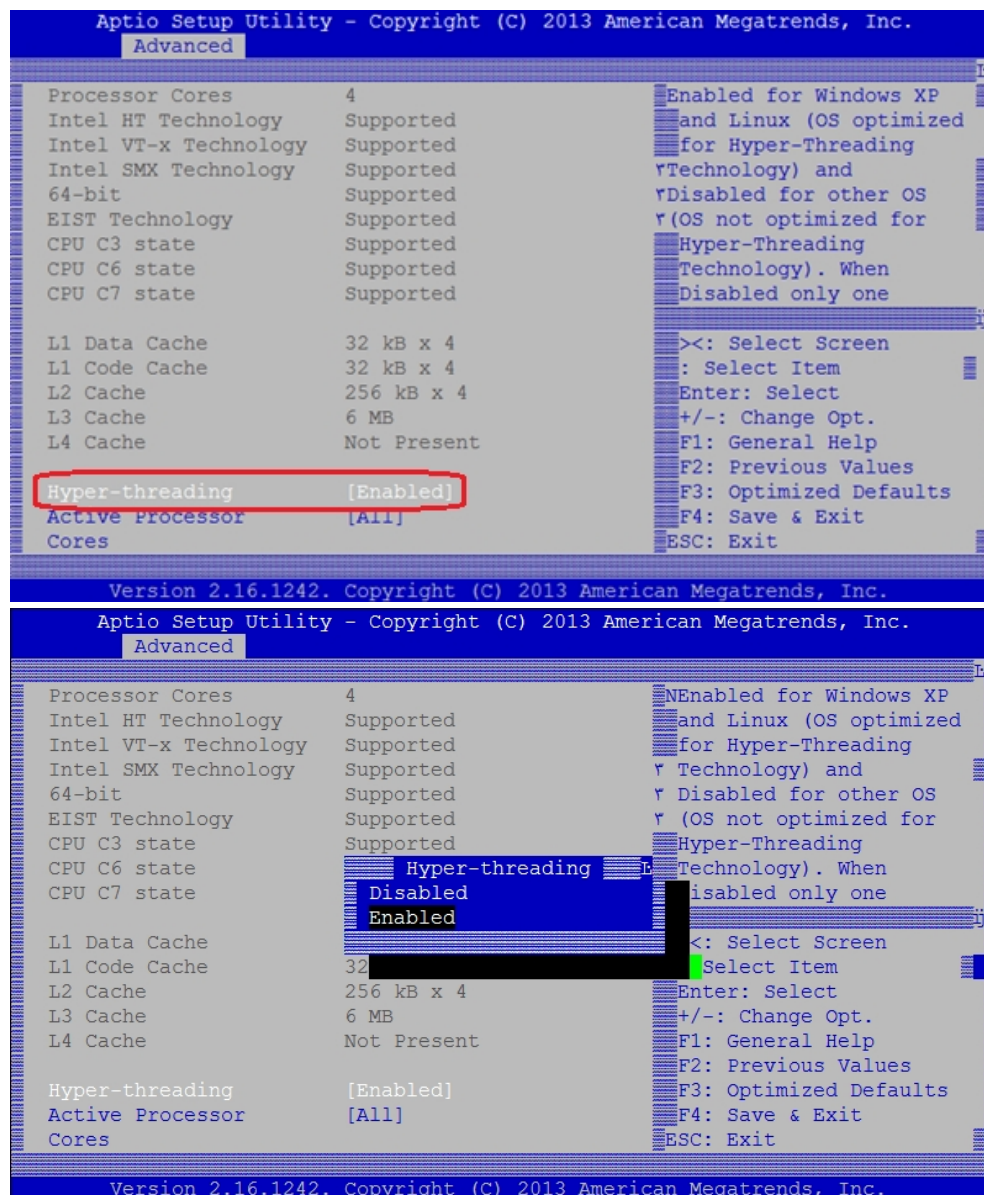


Figure 7-11: Enabling / Disabling BIOS Hyper-threading

Controlling BIST Behavior

The *Built In Self Test* (BIST) feature executes near the end of the BIOS boot process and will indicate if a component has failed, or has otherwise been improperly configured, so that the module will not function as intended.

Note: For additional information on BIST, refer to Appendix E - *BIST Device Information* on page 105.

BIST can be configured to either allow the module to attempt to continue the boot process, or it can halt the module to wait for the user to see the error message and press a key to continue.

To Enable / Disable BIST:

1. In the BIOS Menu, select the Advanced tab, and then select *BIST Configuration* (Figure 7-12 below)

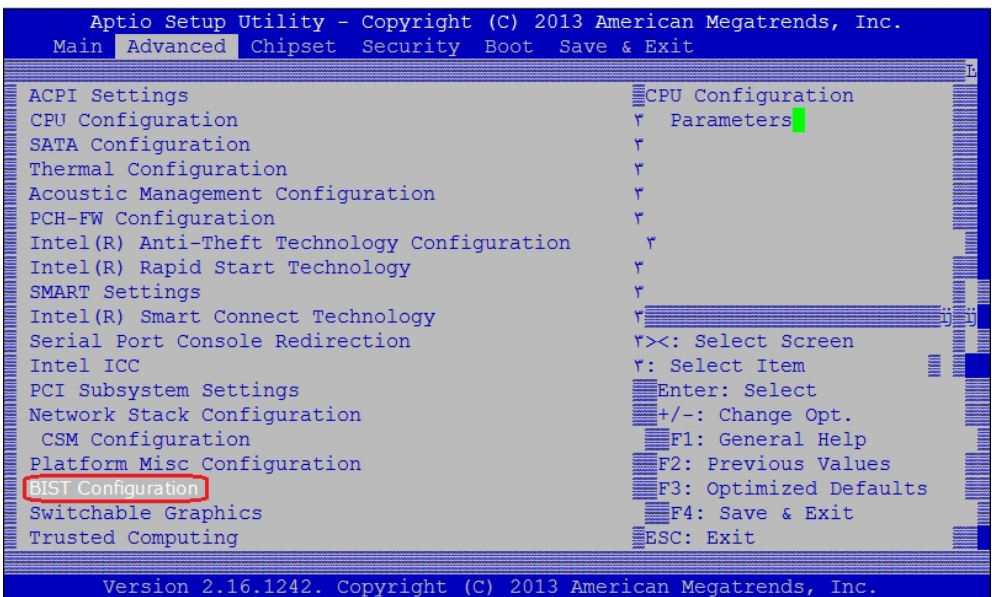


Figure 7-12: Selecting the BIST Configuration option

2. On the *BIST Configuration* screen (Figure 7-13 on the facing page).
3. Select *None* to *disable* BIST, or select another option to *enable* BIST.

Note: The BIST level options are summarized, below, in Table 7-1 on page 90

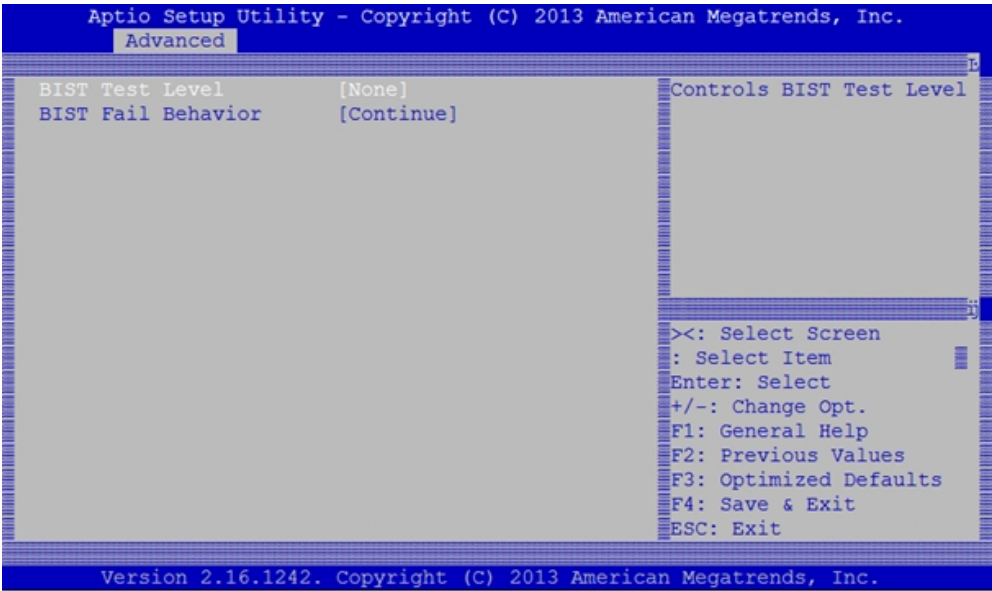


Figure 7-13: Selecting BIST Test Level options

4. Select *BIST Test Level* to display the *Test Level options pop-up* (Figure 7-14 below). The level options are listed in the following table (Table 7-1 on the next page).

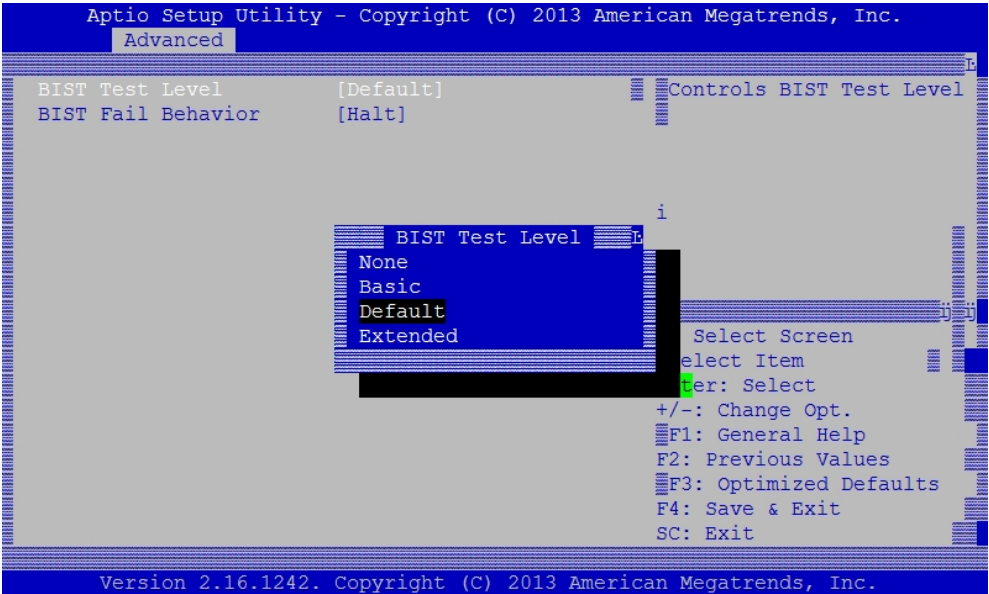


Figure 7-14: BIST Test Level options

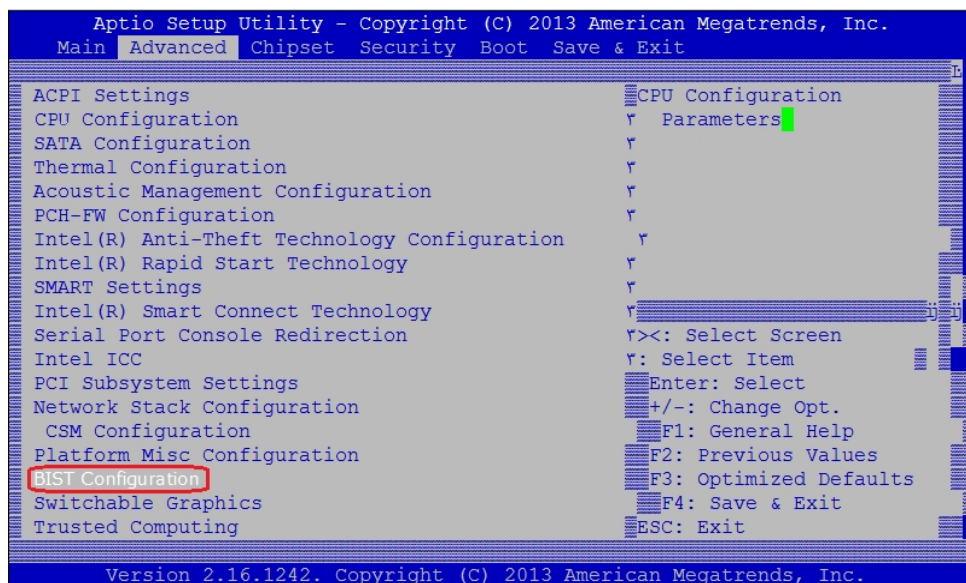
Table 7-1: Selecting BIST Level

Level	Time (Sec)	Tests Performed
None	0	No BIOS self-tests executed upon power-up.
Basic	<2	Validates the link status on all accessible PCIe and Ethernet interfaces. Validates basic connectivity to all accessible devices.
Default	<20	Includes all Basic level tests. Validates functionality on accessible devices.
Extended	<120	Includes all Default level tests, as well as extensive memory testing.

Changing BIST Fail Behavior

To Change BIST Fail Behavior:

1. In the BIOS Menu, select the Advanced tab, and then select BIST Configuration(Figure 7-15 below).

**Figure 7-15:** Configuring BIST Fail behavior

2. On the BIST Configuration screen, select *BIST Fail Behavior*(Figure 7-16 on the facing page).

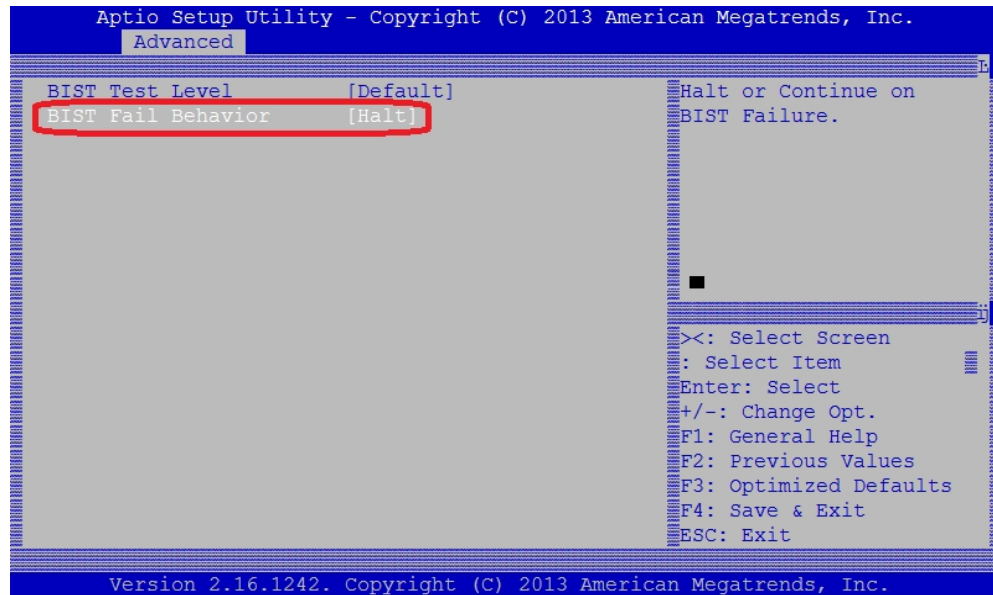


Figure 7-16: BIST Fail Behavior option

3. At the BIT Fail Behavior options pop-up (Figure 7-17 below), select *Halt* to stop the BIST process, or *Continue* to continue booting if BIST detects any errors during BIST execution at POST time.

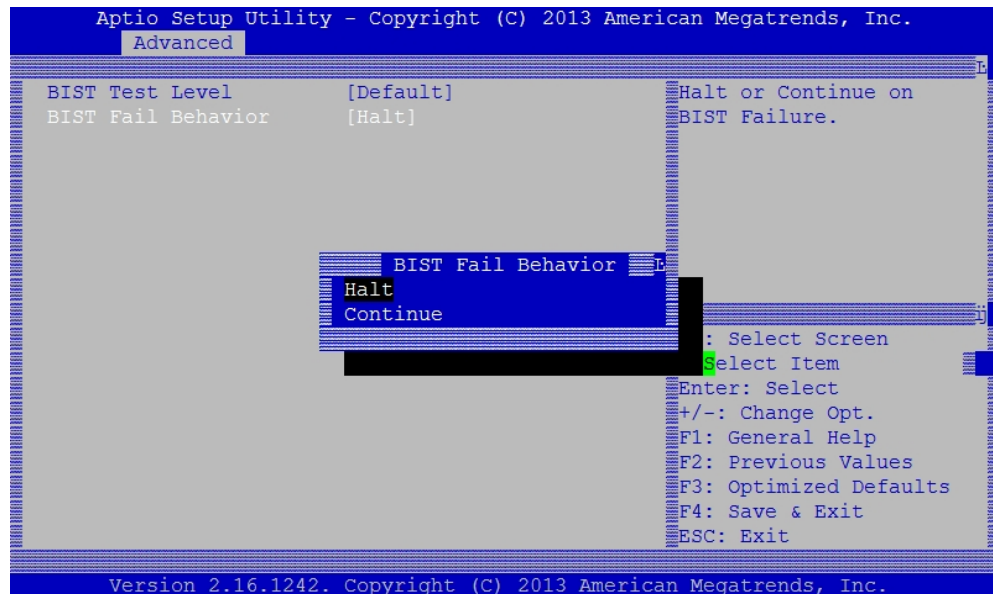


Figure 7-17: Selecting BIST Fail *Halt* or *Continue* option

Configuring PCIe Backplane Resets

The PCIe Backplane Resets feature allows the user to control whether on-board PCIe bus resets are propagated through the backplane connectors. There are two reset controls, *PCIe Backplane Reset 1* and *PCIe Backplane Reset 2*, which are wired to the backplane expansion connectors P5 and P2, respectively.

- When the reset control for a given reset line is *disabled* (the default), then a PCIe bus reset on the LDS6525 will *not* be propagated out to the associated backplane expansion bus.
- When the reset control for a given reset is *enabled*, then a PCIe bus reset on the LDS6525 *will* be propagated out to the associated backplane expansion bus, and to any PCIe devices located on that expansion bus.

To configure the PCIe Backplane Resets:

1. In the BIOS Menu, select the Advanced tab, and then select *Backplane PCIe Reset Controls* (Figure 7-18 below).

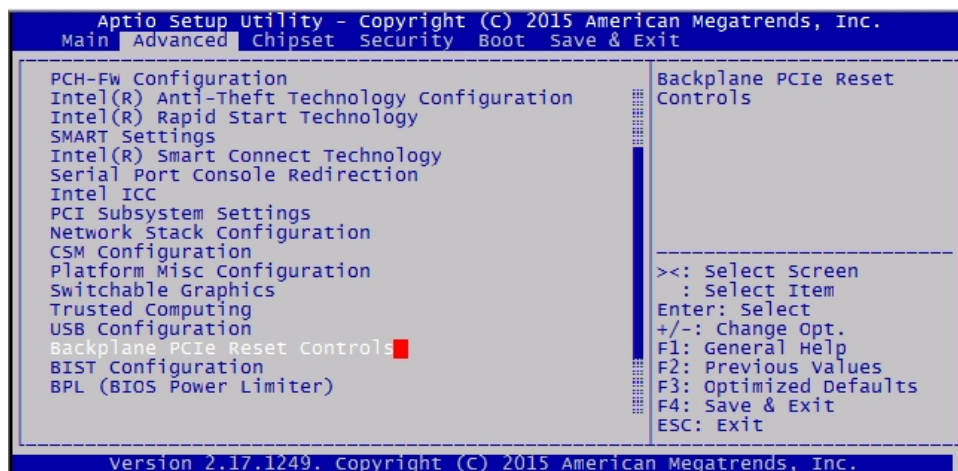


Figure 7-18: Selecting the PCIe Backplane Configuration option

2. In the Reset Enabled/Disabled screen, at right, use the up and down arrows to select the *Select Item* option, and select the port reset number applicable to your backplane configuration to display the *Disabled/Enabled* pop-up menu (Figure 7-19 on the facing page).
3. Using the up/down arrows, select *Disabled* or *Enabled* to disable or enable the reset function for the associated backplane.

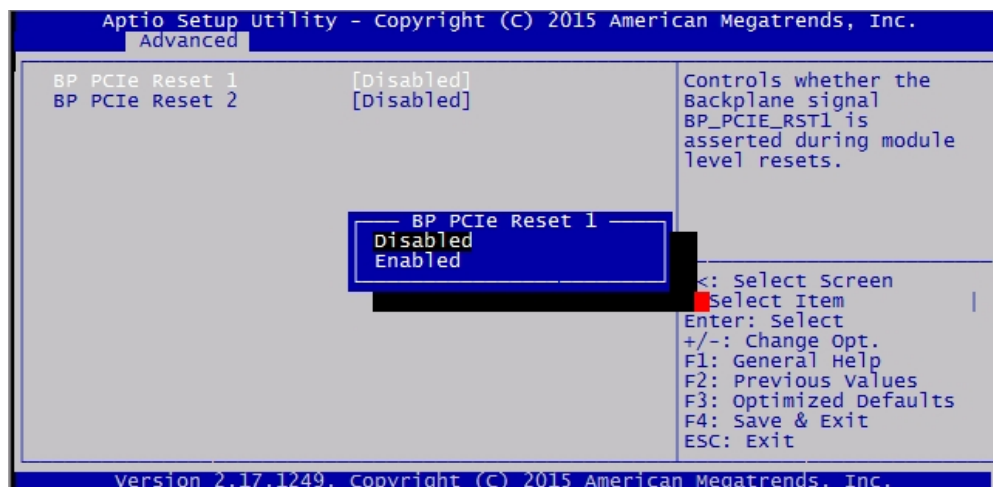


Figure 7-19: Selecting the BP PCIe Reset *Disabled/Enabled* option

4. After making your selection, press the <F4> key, to save your selection and exit the BIOS utility.

Configuring BPL (BIOS Power Limiter)DOC-DVG-LDS6525-V1.0

Use the BPL feature to control CPU clock speed minimums, maximums, and boot speeds. Set the minimum and maximum cpu speeds advertised to the system, ranging from the platform lower/upper limits (including TurboMode), or any step values within the platform supported ranges.

Example: If the board has a 1.8Ghz cpu populated, which supports cpu speeds from 800Mhz to 1.8Ghz (with 2.1Ghz max Turbo), the user may set the “minimum” speed to any 100Mhz-step boundary from 800Mhz up to 1.8Ghz. The user may set the “maximum” speed to any 100Mhz-step boundary from 800Mhz up to 1.8Ghz, *or* may set the speed to 1.9Ghz which would indicate that Turbo should be enabled. Additional limits are that the “max” speed selected must be equal or greater than the “min” speed selected. If invalid values are selected, the BIOS will adjust the min/max speeds to force them to be valid values.

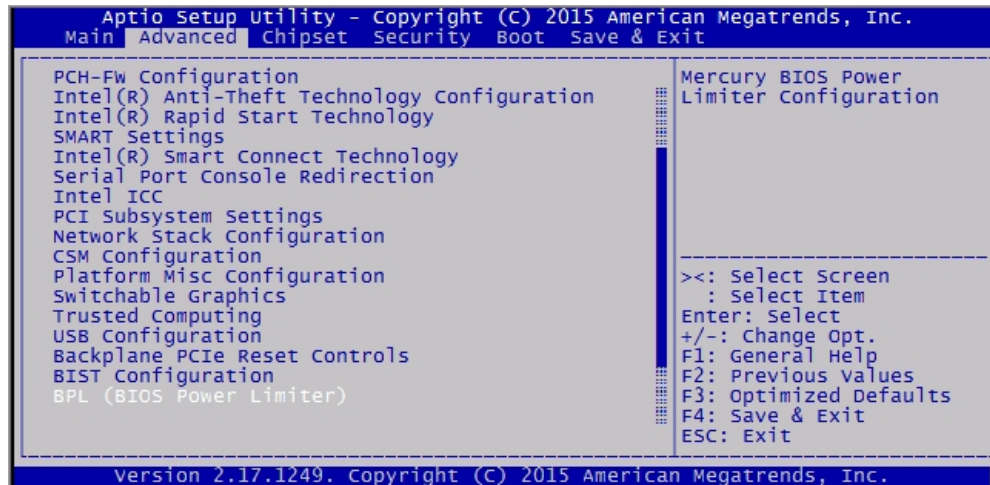
If a user selects the same minimum and maximum value, then the cpu will be “locked” at the fixed value chosen, and an acpi-compliant OS will not be allowed to change the cpu clock speed at runtime.

All cpu cores within the system share the same BPL settings.

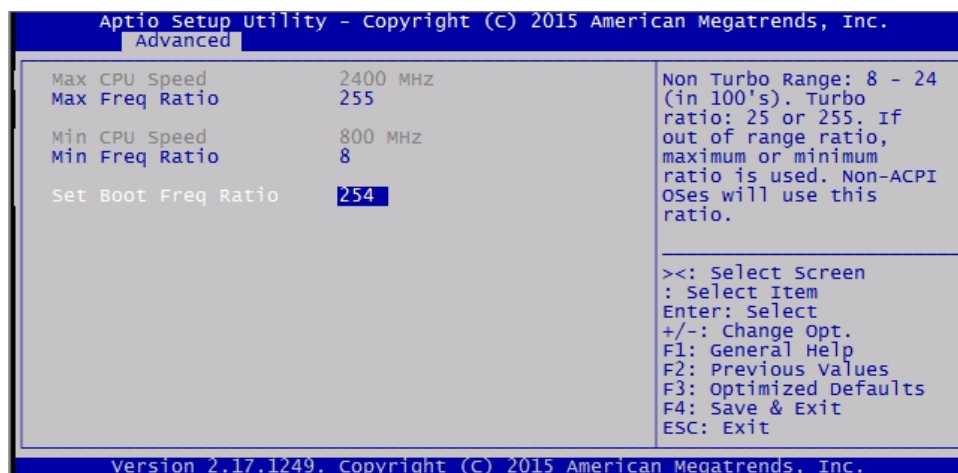
The “boot” speed can be selected independently of the minimum and maximum speeds. It’s important to note that most current OS’s are acpi-aware, so the OS can choose to select a new CPU clock speed shortly after booting, if the minimum/maximum BPL values allow it. A legacy OS (or an OS with ACPI support disabled) will not change from the selected CPU boot speed.

To Configure the BIOS Power Limiter:

1. In the BIOS Menu, select the Advanced tab, and then select *BPL (BIOS Power Limiter)*, as shown in Figure 7-20 below.

**Figure 7-20:** Selecting the BPL option

2. Use the up and down arrows to select from *Max/Min CPU Speed and Frequency Ratio*, and *Boot Frequency Ratio* parameters (Figure 7-21 below).
3. Press <Enter> to select the parameter for editing.

**Figure 7-21:** Editing BPL parameters

4. Edit the BPL parameters as needed for your configuration requirements, and press the <F4> key to save and exit the BIOS utility.

Upgrading the BIOS Firmware

Upgrades to the LDS6525 module BIOS are performed using Flash updates. To determine if there is a BIOS update available, contact Mercury Global Support Services.

The following packages are required to perform flash updates. The version number can vary depending on the release.

- **mrchy-ami-flash-utility** - The mrchy-ami-flash-utility RPM installs the necessary utilities from AMI to update a flash image. The utility for updating the flash is stored at the location:
`/mercury/frs/LDS6525/bios/AMI.`
- **mrchy-frs-lds6525** - The mrchy-frs-lds6525 package installs the BIOS image at the location: `/mercury/frs/LDS6525/bios/LDS6525_base_bios_vxxx.bin`

To perform a flash update:

1. Open a terminal window, and log in as root.
2. At the command prompt, enter the following command at hit <Enter>:

```
# cd /mercury/frs/LDS6525/bios/AMI
```

3. At the command prompt, enter the following command at hit <Enter>:

```
# ./afulx_64 /mercury/frs/LDS6525/bios/LDS6525_base_bios_vxxx.bin /P  
/B /REBOOT
```

Note:

- In the event that the module does not boot and recovery mode is required, contact Mercury Global Support Services
- The user also has the option of using “frs_load” to update the BIOS. This tool’s usage is detailed in the LDS6525 Configuration Guide.

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Appendix A

ESD Precautions

ESD damage can occur when you handle electronic devices. If the devices include printed circuit boards or semiconductor components, ESD can damage the devices, rendering them inoperable.

To prevent ESD damage, follow these standard guidelines:

- Work in a static-free area.
- If you must handle a component before installation:
 - Do not place it on a surface that produces ESD (carpeting, for example).
 - Do not place it near devices that are conductive or that create static electricity.
- Never handle electronic components unless you are properly grounded:
 - Always attach a wrist grounding strap before handling electronic components.
 - Always use a commercial wrist grounding-strap, not a homemade device.
 - Ensure that the strap makes direct (maximum) contact with bare skin.
 - The other end of the wrist strap must make contact with a suitable ground.
 - Use the chassis ESD/ground port. If the chassis does not have such a port, connect the wrist strap to any exposed metal surface on the chassis.
- Do not remove an electronic device from its protective package until immediately before you install it.
- Immediately before removing the protective material from an electronic device, touch the protective material to the chassis into which the device will be installed.
- Minimize motion when handling unpackaged electronic devices. Otherwise harmless motion, such as two pieces of fabric making contact, can generate static electricity sufficient to damage an electronic device.
- To avoid contact between an electronic device and your clothing, wear an ESD smock. ESD emitted from clothing can damage components.

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- After removal, always place the component on a grounded, static-free surface or ESD pad, or inside a proper ESD bag.
- Always place boards component-side up on an antistatic surface, in an anti-static card rack, or in an electrostatic bag.

Appendix **B**

MTBF Realism

More than a hundred industry standards are available to predict MTBF, but none accurately predict the MTBF realism. A realistic MTBF is understood to be the basic MTBF for the purposes of estimating the field return rate of the field replaceable unit (FRU).

Basic MTBF is only for FRU hardware failures and it intentionally avoids the system fault tolerance for redundancy. For example, if a system has multiple redundant power supply FRUs, the realistic basic MTBF is the average rate of returns for the power supply FRU in that system application.

It is feasible that the average return rate of 500,000 hours is observed while the average system MTBF for the redundant power supply function is well over 10,000,000 hours MTBF.

The observed basic MTBF is influenced by producer, customer and maintenance factors that were not considered in the MTBF prediction standards. Some of the unspecified producer factors include, but are not limited to:

- Hardware design strength
- Software design strength
- Manufacturing strength
- Environmental stress strength by the manufacturer
- Quality of procured parts

Some of the unspecified customer factors include, but are not limited to:

- Environmental stress strength by the customer
- Customer use conditions
- Maintenance practices

Mercury Systems, Inc. has some observed MTBF realism for some high-usage products that were used in typical comfort-controlled user applications. The Mercury-observed realistic MTBF data was approximately 2.5 times the predicted MTBF.

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Appendix C

Example FRU Information

The following is an example of the LDS6525 FRU information. For more information, see the *Ensemble 6000 Configuration Guide*.

```
[Board Info]
Version = 1
Language Code = en
Mfg Date/Time = 03/25/2014
Manufacturer = Mercury Systems, Inc.
Product Name = LDS6525.....
Serial Number = MHG-00100-B2....
Part Number = 420-1231B1-04...
FRU Programmer File ID = fru-info.inf
Custom 1 = 100.....

[Product Info]
Version = 1
Language Code = en
Manufacturer = Mercury Systems, Inc.
Product Name = LDS6525.....
Part/Model Number = 910-56000-01....
Product Version = 100.....
Serial Number = MHG-00104-B2....
Asset Tag = MCS000000.....
FRU File ID = fru-info.inf
```

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Appendix **D**

Flash Partitioning

Figure D-1 below shows the flash memory map for the LDS6525.

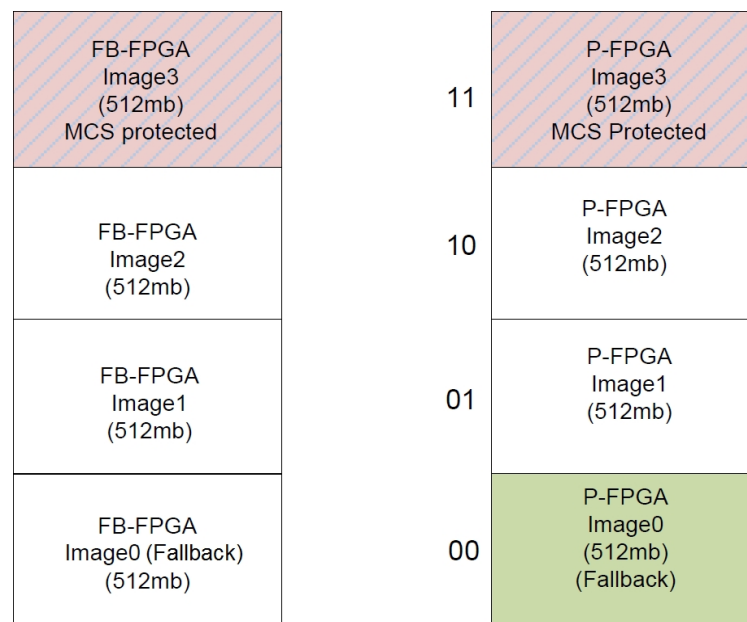


Figure D-1: FB-FPGA / P-FPGA Flash Memory Map for the LDS6525.

There are two FPGA configuration Flash devices on the LDS6525. One device is used to store the FB-FPGA images and the second device stores the P-FPGA images. The flash is symmetrically organized with 128 Kbyte array blocks, has an initial access time of 100 ns and supports synchronous burst reading. The flash is write protected by gating the write signal with the LOCAL_WP_N signal, the MCS_WP_N, and the address range being accessed.

Flash Protection

The Flash allocation is protected as follows:

- Quadrants 0 -3: The assertion of the LOCAL_WP_N signal will disable all modification to the content of this region.

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- Quadrants 3: Both the MCS_WP_N and the LOCAL_WP_N signals will disable all modification to the content of this region.

None of the gating logic can reside in the P-FPGA. Fallback reconfiguration is supported on the Ultrascale device. To support a customer modifiable image and a fallback image for the P-FPGA, flash memory will be allocated as shown above. Note that fallback reconfiguration only works if AES encryption is disabled. When AES encryption is enabled, only a single configuration cycle is attempted; if this cycle fails, the P-FPGA will not be configured unless some external action is taken.

Appendix **E**

BIST Device Information

Overview

Built-in self test (BIST) information for the LDS6525 includes:

- BIST test execution times
- Functional block tests performed by the BIST tool

For information on the BIST tool, see the *Mercury System Management Using SNMP* technical note.

BIST Execution Times

Table E-1 below lists the BIST test execution times for the LDS6525.

Table E-1: BIST Execution Times on the LDS6525

Test Level	Execution Time
None	less than 1 second
Basic	less than 10 seconds
Default	less than 20 seconds
Extended	less than 60 seconds

Functional Block Tests

The functional block tests performed by the BIST include:

- MEM Functional Block
- PROC Functional Block
- PCIe® Functional Block
- Ethernet (ETH) Functional Block
- Management Processor Functional Block

MEM Functional Block Tests

The Basic, Default, and Extended test levels validate the address and data bus interfaces to DRAM.

PROC Functional Block Tests

The Basic, Default, and Extended test levels validate processor to PCH connectivity and on-board NAND access.

PCIe Functional Block Tests

The Basic, Default, and Extended test levels validate the links between the Processor and PCIe switch A.

ETH Functional Block Tests

The Basic, Default, and Extended test levels validate the PCIe link between the PCH and the on-board LAN device.

Management Processor Functional Block Tests

The Basic, Default, and Extended test levels validate access between the CPU and Management Processor.

Appendix **F**

Getting Assistance

Mercury's comprehensive customer services provide support throughout the full product life cycle. The Mercury team can address your requirements from pre-installation through end-of-life. This includes:

- Technical assistance
- Returning items for repair
- Product training

To obtain technical assistance

You can contact the Mercury Global Support Services team as follows:

New Customers	Returning Customers
1. Visit http://www.mrcy.com/. 2. Select Support > Support Overview. 3. On the Flexible Support Offerings page, click the Login Request link. 4. Complete the online form.	1. Visit http://www.mrcy.com/. 2. Select Support > Support Overview. 3. Use any of the links available to open the Secure Customer Login page.

Other Contact Options

Email*	Telephone	Fax*
support@mrcy.com	US 800-872-0040 International 1-978-967-1400	978-256-3558

* When sending a fax or email, include your name and your company name.

The Mercury Global Support Services page also provides Technical documentation and other information resources.

To submit a return request

To submit a return request online, you can contact the Mercury Global Support Services team as follows:

New or Returning Customers
1. Visit http://www.mrcy.com/. 2. Select Support > Support Overview. 3. Under the RMA Request tab, click the RMA Request link. 4. Complete the online form.

Other Contact Options

Email*	Telephone	Fax*
support@mrcy.com	US 800-872-0040 International 1-978-967-1400	978-256-3558

* When sending a fax or email, include your name and your company name.

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